Invoice-Currency Mismatch, Firm Cash Flows, and the Real Effects of Exchange Rates

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Abstract

In the short run, exchange rates move the value of trade far more than its volume. With prices set in the invoicing currency and quantities slow to adjust, a firm's currency exposure depends on the currency of its trade invoices, not on that of its trading partners. Using French customs and balance-sheet data, I construct a shift-share index of each firm's net foreign-currency invoicing position, identified from dollar-invoiced trade between non-U.S. parties. This index outperforms trade-weighted effective exchange rate indices in explaining cash flows of trading firms. Half to two-thirds of the valuation shock reaches cash flow, but firms hold about 70 percent of it as cash and working capital. Real effects concentrate on small, financially constrained firms. The aggregate response stays muted because small exporters are operationally hedged and small importers account for little of total trade.

JEL classification: F14, F31, F41, G32.
Keywords: exchange rates, invoicing currency, cash flow, pass-through, valuation effects

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1 Introduction

International prices are sticky in the currency of invoicing, most often the dollar. The trade literature draws one main consequence from this. Because an exchange-rate movement changes relative prices little, trade volumes barely respond. This is a successful account of why trade volumes look disconnected from exchange rates in the short run. A second consequence has drawn less attention. Besides holding prices and quantities fixed, invoice price stability revalues the trade flow in domestic-currency terms. A shipment priced in a foreign currency is worth more when the domestic currency depreciates. This paper asks whether these valuation effects are present, how large they are, and whether they affect firms' outcomes.

This paper argues that, within the one-year horizon, exchange-rate movements change the value of trade far more than its volume. A firm's exposure is set by the currency in which its trade is priced, rather than by how much it trades or where. This revaluation appears mainly as cash-flow fluctuations in firms' accounts. The paper measures the valuation-induced effect of exchange-rate movements for three outcomes, the value of transactions, the cash flows of trading firms, and their spending. It then asks whether the valuation movements survive aggregation across firms.

I build an illustrative model in which two otherwise identical firms price the same product in different currencies. Under sticky prices, their profits differ to first order, in proportion to the exchange-rate movement. The invoicing literature typically asks *why* firms invoice as they do. It evaluates *ex-ante* profits at the firm's chosen reset price, where the envelope theorem makes the profit difference across invoicing currencies second order.[1] In practice, firms are more likely to inherit yesterday's price, and at that price a payment preset in a foreign currency is revalued one-for-one in domestic-currency terms and passes straight to cash flow.

Ex post, the profit difference across currencies also reflects the quantity response induced by the exchange rate change. Whether valuation *dominates* the quantity response is an empirical question, and it does when the short-run trade elasticity is low enough. I verify this condition in the data. I also outline how a valuation affects firms' input and investment spending through two channels, cheaper internal funds and the expected profitability of an extra unit of

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[1] See, for instance, Engel (2006), Gopinath et al. (2010), Goldberg and Tille (2016), Chung (2016), Amiti et al. (2022), Mukhin (2022), Drenik et al. (2022).

capital. This financing margin governs which firms turn the shock into investment and payroll. Hedging does not remove the valuation-induced internal funds, since hedge maturities are short relative to how long prices stay fixed.

I take this framework to French customs records, which report the invoice currency, unit value, and quantity of every shipment traded with countries outside the European Union. I confirm that invoicing decisions are predetermined and stickier than prices, unchanged over the sample for 99 percent of trade relationship flows. This lets me compare flows that face the same euro movement but are priced in different currencies. The identification relies on a shift-share design. It requires the valuation shocks to be exogenous but not the currency choice to be randomly assigned (Borusyak et al. 2022).

I find that foreign-invoiced border prices (expressed in euro terms) respond strongly to a euro depreciation while euro-invoiced prices barely move, a well-known pass-through asymmetry in the literature (Gopinath et al. 2010). A less emphasized fact is that volumes move less than prices. Measured against the price the buyer faces, the trade elasticity conditional on the exchange-rate shock is about 0.3 to 0.4, similar across invoicing regimes. Since quantity moves less than the price, a domestic currency depreciation primarily increases the *value* of the trade flow, from the point of view of the domestic firm. Within the year, a foreign-priced flow then acts like a foreign-currency-denominated financial position.

I then aggregate foreign-priced operations to the firm and build a shift-share index of invoice-currency mismatch. Its shares are lagged exposures interacted with the exchange-rate movement. The index accounting for the invoice currency of trade operations outperforms a standard trade-weighted effective exchange rate in a horserace, raising the estimated cash-flow response by a factor of three to five. It also absorbs the explanatory content of the trade-weighted measure. This resolves a tension in the firm-exposure literature, where regressions on trade-weighted exchange rates yield small, unstable exposures (Jorion 1990, Bartram et al. 2010, Dominguez and Tesar 2006).

Cash flow moves with invoice-currency mismatch at a magnitude comparable to the revaluation measured at the border, 67 cents per euro of exposure. This response has little effect on real decisions on average, with no employment or investment effect and only a limited salary effect. The average firm holds the cash flow as liquidity rather than spending it, and about 70 percent of the cash flow accumulates as cash and net working capital. Behind this average, firms differ in how they use the cash flow. Smaller and more financially constrained firms spend more of it on salaries and, at the margin, investment. Firms with ample internal funds retain it. The same pattern appears whether internal funds are proxied by size, listing, or a financial-constraint index, though no single proxy can sharply identify a financial constraint vs. a scale-related marginal profitability (as in q-theory).

These firm-level regressions condition on each firm's exposure. They identify the response to one unit of invoice-currency mismatch, not the amount of mismatch a firm carries. Whether the valuation effect aggregates is a separate question, which depends on how exposure is distributed across firms as well as on its average transmission. This distribution has three features. First, net importers are short the dollar across the entire size distribution, most intensely among the smallest, where the net position reaches about 14 percent of sales. Second, within-firm natural hedging, the netting of dollar imports against dollar exports, offsets only a fraction of exposure and only for some firms. It absorbs about a quarter of aggregate dollar exposure and is mainly how small and medium exporters neutralize their positions. Third, most aggregate exposure survives the firm-level natural hedging, about three-quarters of aggregate dollar trade, and it is highly concentrated. The top 1 percent of firms by trade value hold 94 percent of the long positions and 70 percent of the short positions.

At the aggregate level, a dollar appreciation raises net exporters' cash flow and lowers that of net importers. In France, the aggregate exposures of net exporters and net importers are roughly equal and opposite, so their real responses cancel. In addition, even within each exposure position the lion's share of the exposure is held by the firms least likely to spend it. Among both net exporters and net importers, the foreign-currency exposure is held by large, liquid firms whose propensity to spend is near zero. The high-propensity firms are either operationally hedged, as with small and medium exporters, or too small in nominal terms for their spending to aggregate, as with small net importers. A 10 percent dollar appreciation raises net exporters' cash flow by about €2.94bn and lowers net importers' cash flow by about €2.6bn, each roughly 0.1 percent of GDP. The two nearly cancel, and the net cash-flow effect is €0.37bn, about 0.02 percent of GDP.

A near-zero aggregate effect can still hide substantial firm-level effects. The clearest case is small net importers, which are more likely to be financially constrained and are intensely short the dollar. After an invoice-induced valuation shock they cut payroll and, at the margin, investment, and together they employ about a third of France's trading workforce. Large net exporters, by contrast, hold large cash-flow changes as liquidity. But a muted real response does not mean there is no cost. Carrying large liquidity buffers or hedging the position is itself costly, in ways beyond this paper's scope.

This paper makes three contributions. First, I bring the view that a short-run exchange-rate movement acts mainly as a valuation effect. A firm's exposure then follows from the currency its operations are priced in, and the exchange-rate shock reaches the firm through its cash flow. Second, I measure invoice-currency mismatch at the transaction level and identify its effect from vehicle-currency dollar variation in a shift-share design. This recovers a large and stable exposure that trade-weighted exchange rates miss (Jorion 1990, Bartram et al. 2010, Alfaro et al. 2023a). Third, I show that whether the shock matters in aggregate depends on which firms hold the exposure and whether they spend it. In France, dollar positions cancel between net importers and net exporters. What survives sits with large firms that spend little of it, or with firms too small to move the aggregate. The design has since been applied to more exposed economies. For instance, it recovers larger real effects in Thailand, where exporters cut investment and hiring after adverse dollar valuation shocks (Nookhwun et al. 2025).

The remainder of the paper proceeds as follows. Section 2 places these contributions in the related literature. Section 3 develops the model. Section 4 describes the data and institutional setting. Section 5 establishes the border results. Section 6 covers the firm-level cash-flow and real effects and the distribution of exposure across firms. Section 7 documents who holds the aggregate exposure and measures its incidence. Section 8 concludes.

2 Related Literature

This work relates to three literatures on dominant-currency pricing, firm-level exchange-rate exposure, and exchange-rate disconnect. This paper is the first to combine three steps. It measures each firm's invoice-currency mismatch at the transaction level. It traces that mismatch from the border value of trade to cash flow and spending. And it asks how the incidence of that exposure across firms shapes the aggregate.

A large literature documents that international prices are sticky in the invoice currency, most often the dollar (Gopinath et al. 2010, Goldberg and Tille 2008, Gopinath 2015, Gopinath et al. 2020, Boz et al. 2022). Pass-through into destination prices is high for foreign-priced trade and low otherwise, a pattern documented across several economies (Amiti et al. 2014, 2022, Corsetti et al. 2022, Auer et al. 2021, Chen et al. 2022, Benguria and Wagner 2024, Devereux et al. 2025) and surveyed by Gopinath and Itskhoki (2022). This paper studies a different object, the domestic-currency *value* of operations preset in foreign currency, which is what reaches the firm's accounts.

A second literature estimates how exchange rates reach firm outcomes, and has generally found the link weak. Regressions of firm outcomes on trade-weighted effective exchange rates yield modest and unstable exposures (Jorion 1990, Bartram et al. 2010, Dominguez and Tesar 2006, Campa and Goldberg 1995, Nucci and Pozzolo 2001, Ekholm et al. 2012, Alfaro et al. 2023a). This paper differs by measuring exposure through invoicing rather than trade per se. The closest contemporaneous papers (Adams and Verdelhan 2025, Welch and Zhou 2024) also recover larger exposures by improving exchange rate exposure measurement, but for listed firms and from financial data. Adams and Verdelhan (2025) read transaction and translation risk from the disclosures of public firms, and Welch and Zhou (2024) re-estimate the exchange-rate sensitivity of publicly traded U.S. corporations from their returns.

A third literature studies exchange-rate disconnect. Exchange rates originate largely in financial markets and move with little relation to fundamentals (Meese and Rogoff 1983, Obstfeld and Rogoff 2000, Itskhoki and Mukhin 2021, Lilley et al. 2022, Engel and Wu 2023, Kekre and Lenel 2024, Itskhoki and Mukhin 2025). Currency choice is itself an equilibrium that sustains dollar dominance (Mukhin 2022, Gopinath and Stein 2021, Chahrour and Valchev 2022, Drenik et al. 2022, Eren and Malamud 2022, Xie 2025). This paper adds a related step, showing that short-term disconnect lies in how invoice-currency exposure is distributed across firms with different incidence. At the country level, this is the trade-account counterpart of the external-balance-sheet valuation effects in Allen and Juvenal (2025). It also provides the firm-level foundation for the dollar-invoicing transmission estimated in aggregate data (Georgiadis and Schumann 2021, Cook and Patel 2023, Obstfeld and Zhou 2022).

The invoice currency mismatch studied in this paper is unlike the balance-sheet currency mismatch usually associated with debt denominated in foreign currency (Céspedes et al. 2004, Galindo et al. 2003, Calvo et al. 2004, Salomao and Varela 2022, Kalemli-Özcan et al. 2021, Bruno and Shin 2023). Closest in setting, Berthou et al. (2022) use the same French data to study export volumes under dollar invoicing. This paper instead studies how the valuations enter firm accounts and spending. The cash-flow result that is the focus of this paper has since entered the dominant-currency-pricing literature (Gopinath and Itskhoki 2022). A similar design has been reused for Thailand (Nookhwun et al. 2025), the UK after Brexit (Garofalo et al. 2024), and Chile (Cristoforoni and Errico 2024).

3 Model

Under sticky prices a firm typically carries yesterday's nominal price into today's state. At the inherited price, an exchange-rate movement generates differential pass-through and a profit wedge across invoicing regimes.[2] This section maps this wedge into firm value, shows why short-maturity hedging does not undo it, and derives the assumptions needed for the wedge to reach investment and employment. Proofs and supporting derivations are in Appendix D.

[2] The relevant object here is the profit realized at the inherited price, distinct from the reset-price comparison that governs the currency-*choice* problem. The envelope logic that makes the choice second order does not apply to it. Xie (2025) likewise obtains a first-order exposure to exchange rates, through a different channel, in which quantities fixed in advance interact with imperfect financial hedging, so that currency choice carries risk premia ex ante. The wedge here arises ex post from the inherited price itself. It requires neither risk aversion nor hedging frictions, and is present even for a risk-neutral firm with flexible quantities.

3.1 Price setting and firm value conditional on invoicing currency

The first part of this model is similar to the partial equilibrium framework of Gopinath et al. (2010). A French firm exports outside of the European Union and has the following profit function at date t:

$$\Pi(p_t^*; \mathbf{e}_t, \boldsymbol{\eta}_t) \ ,$$

where p_t^* is the log buyer-currency price (expressed in buyer-currency units $*$),

$$\mathbf{e}_t \equiv \left(e_t^{€/*}, e_t^{€/\$} \right)' \tag{1}$$

is the vector of relevant log exchange-rate legs, and $\boldsymbol{\eta}_t$ collects non-exchange-rate demand conditions, cost shocks, and competitor prices. The notation $e_t^{a/b}$ denotes the log price of currency b in units of currency a. Also, let y_t^a denote a variable y expressed in units of currency a at time t.

For notational economy, the profit function suppresses firm primitives such as demand elasticity, markup, imported-input intensity, and the currency composition of marginal costs. The special case in Section 3.2 will make their role more explicit.

Under flexible prices, let $\tilde{p}_t^*$ denote the desired buyer-currency price:

$$\tilde{p}_t^*(\mathbf{e}_t, \boldsymbol{\eta}_t) \in \arg\max_{p^*} \Pi(p^*; \mathbf{e}_t, \boldsymbol{\eta}_t). \tag{2}$$

Since $p_t^j + e_t^{*/j} = p_t^*$, the desired price in currency j is

$$\tilde{p}_t^j = \tilde{p}_t^* - e_t^{*/j} = \tilde{p}_t^* + e_t^{€/*} - e_t^{€/j}. \tag{3}$$

Price setting. Assume that the firm can adjust its price each period with exogenous probability $1 - \theta$, as in Calvo (1983), but cannot switch its invoicing currency j. Following the standard posted-price representation in Gopinath et al. (2010), let

$$V_j(p^j; \mathbf{e}_t, \boldsymbol{\eta}_t)$$

denote the value of a firm invoicing in currency j when the currently effective posted price in that currency is p^j.

The value of the firm with currently effective posted price p^j satisfies

$$\begin{aligned} V_j(p^j; \mathbf{e}_t, \boldsymbol{\eta}_t) = {} & \Pi(p^j + e_t^{*/j}; \mathbf{e}_t, \boldsymbol{\eta}_t) \\ & + \beta\theta \mathbb{E}_t \left[V_j(p^j; \mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1}) \right] + \beta(1-\theta) \mathbb{E}_t \left[\bar{V}_j(\mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1}) \right] , \end{aligned} \tag{4}$$

where $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathbf{e}_t, \boldsymbol{\eta}_t]$, and

$$\bar{V}_j(\mathbf{e}_t, \boldsymbol{\eta}_t) = \max_{p^j} V_j(p^j; \mathbf{e}_t, \boldsymbol{\eta}_t) \tag{5}$$

is the value conditional on resetting the price at date t. The following standard reset-price result fixes the reset price.

Lemma 1 (Reset prices in buyer currency)**.** *To a first-order approximation, the optimal reset price in currency j satisfies*

$$\bar{p}_t^j = (1-\beta\theta)\sum_{h=0}^{\infty}(\beta\theta)^h \mathbb{E}_t\left[\tilde{p}_{t+h}^j\right]. \tag{6}$$

Under a random-walk exchange-rate benchmark all invoicing regimes share the same reset buyer-currency price:

$$\bar{p}_t^j + e_t^{*/j} = \bar{p}_t^* \qquad \textit{to first order,} \tag{7}$$

where $\bar{p}_t^*$ is the reset price under buyer-currency invoicing, (6) *evaluated at $j = *$.*

Because $\bar{V}_j$ in (5) is an optimized value, the envelope theorem makes its first-order component common across invoicing regimes for the same firm, independent of the exchange-rate process. Under the random walk assumption, the reset *prices* coincide as well, not only the values.

We now compare a firm invoicing in currency $j \neq €$ with the same firm counterfactually invoicing in euros. The two firm values are evaluated at the same inherited prices in period $t-1$:

$$p_{t-1}^j + e_{t-1}^{*/j} = p_{t-1}^{€} + e_{t-1}^{*/€} \equiv p_{t-1}^*. \tag{8}$$

Given the pervasiveness of price rigidity, the inherited price is a relevant approximation point for this paper. At any point in time, firms are more likely to report profits at inherited prices than at reset prices.

For any variable y, define the horizon-h log difference relative to the inherited state as

$$\Delta_h y_t \equiv y_{t+h} - y_{t-1}\,, \quad \Delta y_t = y_t - y_{t-1}$$

The random-walk benchmark used below is

$$\mathbb{E}_t[e_{t+h}^{a/b}] = e_t^{a/b} \qquad \text{for all } h \geq 0, \tag{9}$$

so that $\mathbb{E}_t[\Delta_h e_t^{a/b}] = \Delta e_t^{a/b}$ for all $h \geq 0$.

Proposition 1 (Value wedge from an inherited price)**.** *Fix currency of invoicing $j \neq €$. Evaluate the same firm under two output-invoicing regimes, j and €, at inherited prices satisfying* (8)*. Take a first-order approximation around the inherited state*

$$(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}).$$

Then the value wedge from invoicing in currency j vs. euros, where V_j and $V_€$ denote firm value under the two regimes, is, to first order,

$$\begin{aligned} & V_j(p^j_{t-1}; \mathbf{e}_t, \boldsymbol{\eta}_t) - V_€(p^€_{t-1}; \mathbf{e}_t, \boldsymbol{\eta}_t) \\ & \quad = \Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}) \sum_{h=0}^{\infty} (\beta\theta)^h \mathbb{E}_t \left[\Delta_h e^{€/j}_t \right], \end{aligned} \tag{10}$$

where Π_1 denotes the derivative of Π with respect to the buyer-currency price p^*. If exchange rates follow a random walk, then

$$V_j(p^j_{t-1}; \mathbf{e}_t, \boldsymbol{\eta}_t) - V_€(p^€_{t-1}; \mathbf{e}_t, \boldsymbol{\eta}_t) = \frac{\Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})}{1 - \beta\theta} \Delta e^{€/j}_t. \tag{11}$$

The weights $(\beta\theta)^h$ reflect discounting and the probability that the inherited price remains in force through horizon h. The reset component does not appear in (10) because, by Lemma 1, reset values coincide to first order across invoicing regimes.

The first order component of the current-profit wedge $h = 0$ is:

$$\begin{aligned} & \Pi(p^*_{t-1} + \Delta e^{*/j}_t; \mathbf{e}_t, \boldsymbol{\eta}_t) - \Pi(p^*_{t-1} + \Delta e^{*/€}_t; \mathbf{e}_t, \boldsymbol{\eta}_t) \\ & \quad = \Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}) \Delta e^{€/j}_t. \end{aligned} \tag{12}$$

This is the object that maps most directly to observed quantities in the empirical section.

Equation (12) also clarifies the envelope logic that the invoice literature usually considers in evaluating *profit* wedges generated by different invoice currencies. The derivative Π_1 vanishes when it is evaluated at the flexible-price optimum *for the same state*, since by (2), $\Pi_1(\tilde{p}^*_t; \mathbf{e}_t, \boldsymbol{\eta}_t) = 0$. Hence if the inherited price is close to the current desired price, $p^*_{t-1} \simeq \tilde{p}^*_t$, then $\Pi_1(p^*_{t-1}; \mathbf{e}_t, \boldsymbol{\eta}_t) \simeq 0$ and the first-order profit wedge across invoicing regimes is small. Most theoretical papers using first-order logic approximate around the reset price. This leaves open the empirical question of whether first-order exchange-rate effects appear when inherited prices are away from the local envelope point.

The sign of this wedge is itself informative, and it separates the pure valuation effect studied in this paper from the well-known expenditure switching effect. Writing out the two forces makes the condition explicit. Let R_{t-1} be revenues, C_{t-1} be total costs, and $\Pi_{t-1} = R_{t-1} - C_{t-1}$ be the variable profit at the inherited state. Let $\varepsilon \equiv \partial \log Q / \partial p^* < 0$ be its local demand elasticity at the inherited buyer-currency price. To first order, the derivative of the profit function decomposes as

$$\Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}) = \underbrace{R_{t-1}}_{\text{valuation}} + \underbrace{\varepsilon\, \Pi_{t-1}}_{\text{quantity}}. \tag{13}$$

A higher inherited buyer-currency price revalues the preset revenue flow one-for-one. The same higher price moves quantity along demand by ε, and that quantity change scales euro profit per unit, the expenditure-switching term $\varepsilon\, \Pi_{t-1}$. Every price effect therefore splits into a revaluation of the preset flow and a demand response proportional to the elasticity. Whether valuation dominates volume effects in explaining ex-post profit variation is a testable condition, and the empirical sections verify it. The derivation is in Appendix D.3.

3.2 A useful special case and the empirical bridge

This subsection provides a closed-form first-order approximation to profits under different invoicing regimes. It serves two purposes. First, it gives economic content to the derivative Π_1 in Proposition 1. Second, it shows how a firm's exchange-rate exposure depends on observable revenue, cost, markup, demand, and cost-currency shares. This provides an intuitive bridge between the theoretical framework and the econometric specifications used in section 6.

Throughout this section, uppercase variables denote levels and lowercase variables denote logs. Thus log prices are $p = \log P$, log quantities are $q = \log Q$, and log nominal marginal costs are $mc = \log MC$. Profit changes $\Delta\Pi$ are level changes over time expressed in domestic currency (euro, in this context).

Suppose firm i's local demand is

$$Q_{i,t} = D_{i,t} \left(\frac{P^*_{i,t}}{P^*_t} \right)^{-\sigma_i}, \tag{14}$$

where $P^*_{i,t}$ is the buyer-currency price, P^*_t is an aggregate local price index, and $\sigma_i > 0$. Conditional on no reset in output currency j, the inherited buyer-currency price moves one-for-one with $\Delta e^{*/j}_t$. Hence

$$\Delta q^j_{i,t} = -\sigma_i \Delta e^{*/j}_t + \Delta \tilde{d}_{i,t}, \tag{15}$$

where

$$\Delta \tilde{d}_{i,t} \equiv \Delta \log D_{i,t} + \sigma_i \Delta p^*_t \tag{16}$$

collects demand shifters and movements in the aggregate local price index.

Suppose the production function of firm i is constant returns to scale, with Hicksian productivity $a_{i,t}$, home input price w_t, and an imported-input share α_i^k invoiced in currency $k \in \{*, €, \$\}$. Ignoring changes in the foreign-currency price of the imported input itself, the log marginal-cost change in euros is

$$\Delta mc_{i,t} = -\Delta a_{i,t} + (1 - \alpha_i^k)\Delta w_t + \alpha_i^k \Delta e_t^{€/k}. \tag{17}$$

Define lagged level revenue, variable cost, and profit in euros as

$$R_{i,t-1} \equiv P_{i,t-1}^{€} Q_{i,t-1}, \qquad C_{i,t-1} \equiv MC_{i,t-1}^{€} Q_{i,t-1}, \qquad \Pi_{i,t-1} \equiv R_{i,t-1} - C_{i,t-1}. \tag{18}$$

For the same firm evaluated under different output-invoicing regimes, these lagged level objects are common at the comparison point.

Consider first the absolute change in domestic currency profits (in euros) under the j-invoicing regime. A first-order approximation gives

$$\begin{aligned}\Delta\Pi_{i,t}^{j} &\approx R_{i,t-1}\Delta e_t^{€/j} + \Pi_{i,t-1}\Delta q_{i,t}^{j} - C_{i,t-1}\Delta mc_{i,t} \\ &\approx R_{i,t-1}\Delta e_t^{€/j} - \sigma_i \Pi_{i,t-1}\Delta e_t^{*/j} - C_{i,t-1}\Delta mc_{i,t} + \Pi_{i,t-1}\Delta\tilde{d}_{i,t}.\end{aligned} \tag{19}$$

The above condition highlights the importance of controlling for demand effects and for firm productivity effects. Substituting (17), and focusing only on exchange-rate terms, yields

$$\Delta\Pi_{i,t}^{j}\Big|_{\mathrm{FX}} \approx \underbrace{R_{i,t-1}\Delta e_t^{€/j}}_{\text{revenue valuation}} - \underbrace{C_{i,t-1}\alpha_i^k \Delta e_t^{€/k}}_{\text{cost valuation}} - \underbrace{\sigma_i \Pi_{i,t-1}\Delta e_t^{*/j}}_{\text{expenditure switching}}. \tag{20}$$

Revenue valuation loads on the currency in which export revenue is preset. Cost valuation loads on the currency in which imported inputs are preset. The expenditure-switching term loads on the exchange-rate movement that changes the buyer-currency price faced by customers. Therefore, the firm's absolute profit exposure generally depends on both output-invoicing and cost-invoicing choices. In the data, firms that invoice exports in different currencies may also differ in their imported-input shares or in the currency composition of marginal costs. This is why this paper focuses equally on import and export flow exposure.

This also clarifies potential misspecification in analyses that lack invoicing-currency information. Without observing invoice currencies j and k, we must impose a currency for export revenues, a currency for imported input costs, and a currency governing expenditure switching. A common benchmark in the literature assumes producer-currency pricing. If output is priced in euros and imported inputs are priced in foreign currency, then

$$\Delta\Pi_{i,t}^{\mathrm{PCP}} \approx (\sigma_i \Pi_{i,t-1} - C_{i,t-1}\alpha_i^*)\, \Delta e_t^{€/*}. \tag{21}$$

This benchmark has no revenue-valuation term, because export revenues are preset in euros, and it loads all surviving exposure on the single euro/buyer leg $\Delta e_t^{€/*}$. This leg carries two terms that cannot be separated, an expenditure-switching (quantity) term $\sigma_i \Pi_{i,t-1}$ and an imported-input cost-valuation term $-C_{i,t-1}\alpha_i^*$.[3] Dominant-currency pricing, the regime most common in the data and studied in this paper, nests neither,

$$\Delta\Pi_{i,t}^{\mathrm{DCP}} \approx \underbrace{\left(R_{i,t-1} - C_{i,t-1}\alpha_i^{\$}\right)\Delta e_t^{€/\$}}_{\text{net dollar valuation}} - \underbrace{\sigma_i \Pi_{i,t-1} \Delta e_t^{*/\$}}_{\text{expenditure switching}}, \tag{22}$$

so the net dollar position is revalued on the euro/dollar exchange rate leg while expenditure switching acts only on the dollar/buyer currency leg. This paper avoids placing any assumption on the invoice currency of revenues and costs as the customs data reveal the invoice currency directly. In the case of dominant-currency pricing, (22) is taken to the data with the two exchange rate shock legs entered as separate regressors.

3.3 From profits to investment

We now connect the profit and value wedges to firm investment. We write the variable of interest in terms of physical investment and capital, but the same logic applies more broadly to other forward-looking expenditures including inventories, working capital, customer credit, or payroll commitments financed in advance.

Let I_t^j denote investment chosen at the end of period t by a firm invoicing in currency j, and let

$$K_{t+1} = (1-\delta_K)K_t + I_t^j\,. \tag{23}$$

The firm finances investment with internal funds, operating profit Π_t^j plus non-operating resources $\bar{n}_t$, and costly external funds b_t^j:

$$I_t^j = \Pi_t^j + \bar{n}_t + b_t^j, \tag{24}$$

with external-finance cost $\mathcal{C}(b_t^j, K_t)$, where $\mathcal{C}_b > 0$ and $\mathcal{C}_{bb} > 0$ at the expansion point. Because $\bar{n}_t$ is common across invoicing regimes, the moving part of internal funds is operating profit Π_t^j, which is what the responses below track.

Let $V_t^j(K_t, \Pi_t^j; p^j, \mathbf{e}_t, \boldsymbol{\eta}_t)$ denote the value of a firm invoicing in currency j, with inherited posted price p^j, scale K_t, and internal funds tracked by operating profit Π_t^j (the constant $\bar{n}_t$ suppressed). Marginal q is then the marginal continuation value of one more unit of K_{t+1},

[3] Destination-currency pricing is its mirror image, with a revenue-valuation term but no expenditure switching.

derived from that same value function:

$$q_t^j \equiv \beta \mathbb{E}_t \left[V_{K,t+1}^j(K_{t+1}, \Pi_{t+1}^j; p^j, \mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1}) \right]. \tag{25}$$

The investment condition holds:

$$q_t^j - 1 = \mathcal{C}_b(I_t^j - \Pi_t^j - \bar{n}_t, K_t). \tag{26}$$

All response equations below are evaluated at the common inherited-state expansion point, where external funds coincide across regimes and are denoted b_t. To study exchange-rate effects, the relevant objects are deviations from that inherited state. For any level object X_t^j, write

$$\Delta X_t^j \equiv X_t^j(\mathbf{e}_t, \boldsymbol{\eta}_t) - X_t^j(\mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}), \tag{27}$$

and define the same-firm differential response relative to euro invoicing as

$$\Delta X_t^{j-€} \equiv \Delta X_t^j - \Delta X_t^{€}. \tag{28}$$

When focusing only on the exchange-rate effect, the non-exchange-rate components of $\boldsymbol{\eta}_t$ are held fixed.

A first-order expansion of (26) splits the investment response into two channels:

$$\Delta I_t^{j-€} \approx \underbrace{\Delta \Pi_t^{j-€}}_{\text{cash-flow channel}} + \underbrace{\frac{1}{\mathcal{C}_{bb}(b_t, K_t)} \Delta q_t^{j-€}}_{\text{marginal-}q\text{ channel}}. \tag{29}$$

Combining the cash-flow response from proposition 1 and assuming a random walk exchange rate process we can further express it as,

$$\Delta I_t^{j-€} \approx \left[\Pi_1(p_{t-1}^*; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}, K_t) + \frac{1}{\mathcal{C}_{bb}(b_t, K_t)} \frac{\beta\theta}{1-\beta\theta} \Pi_{1K}(p_{t-1}^*; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}, K_t) \right] \Delta e_t^{€/j}. \tag{30}$$

Equation (30) is the investment-response counterpart of the profit result.[4] The first term is the cash-flow channel, where the shock moves internal funds, and the second is the marginal-q channel, present only if the profit wedge varies with scale ($\Pi_{1K} \neq 0$).

Both terms scale the same exchange-rate innovation $\Delta e_t^{€/j}$, so the empirical estimates will

[4] The single euro/j leg here does not drop the quantity response. $\Pi_1 = R_{t-1} + \varepsilon \Pi_{t-1}$ carries both the valuation and the quantity (expenditure-switching) response, and the same-firm differential places both on the euro/j leg because $\Delta e_t^{*/j} - \Delta e_t^{*/€} = \Delta e_t^{€/j}$. The firm regression in the empirical part does not impose this collapse. It enters the valuation and expenditure-switching legs separately, so the estimated investment coefficient identifies the valuation effect holding the volume effect fixed (section 6.2).

recover their combined effect, not each channel on its own. A valuation that increases current cash flow also increases the marginal value of scale and both move forward-looking spending the same way. This is why this paper does not treat the shock as an instrument for current cash flow alone. Investment, inventories, and other outlays paid in advance can also respond through marginal q.

3.4 Short-maturity hedging

A natural objection is that firms can hedge foreign-currency transactions, especially when quantities respond little relative to prices. Alfaro et al. (2023b), using the universe of FX-derivative transactions in Chile, find that firms in international trade hedge their *gross* rather than net currency cash flow. They also find that firms pay higher premiums for longer-maturity contracts, so the hedging that firms buy is partial and concentrated at short horizons.[5]

Hedging can solve short-run transaction risk, not medium-run pricing exposure. A forward contract locks in the euro value of a known future receipt at the forward rate quoted when the contract is written. Suppose a firm holds forwards of maturity up to L when a date-t exchange-rate shock arrives. Cash flows settling within L periods were hedged before the shock, so their euro value is fixed within that period. Cash flows settling at horizons $h \geq L$ can be hedged only from date t onward, once forward rates already incorporate the new exchange-rate level. In the best case for hedging, with every transaction settling within L fully insured, the residual value wedge is

$$V_t^{j-€}(L) \approx \frac{(\beta\theta)^L}{1-\beta\theta}\,\Pi_1(p_{t-1}^*; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})\,\Delta e_t^{€/j}, \tag{31}$$

which differs from the unhedged wedge of Proposition 1 only by the factor $(\beta\theta)^L$. Hedging shrinks the wedge in the hedge horizon and no faster. The Calvo price spell lasts $1/(1-\theta)$ periods on average. When the hedge horizon is short relative to that duration, a first-order wedge remains even after every individual transaction is hedged, because the exposure originates in the inherited price. The full algebra is in Appendix D.7.

[5]Forward liquidity is concentrated at short maturities. In the BIS data, FX swaps and outright forwards are overwhelmingly short-dated. Most turnover is at tenors of a week or less, roughly four-fifths of outstanding FX-derivative notional matures within a year, and liquidity thins markedly beyond one to two years (Bank for International Settlements 2025, Borio et al. 2022).

4 Data and Institutional Setting

4.1 Data sources

The analysis combines French Customs administrative records on export and import transactions outside the European Union from 2000 through 2018 with balance-sheet information of French legal entities from 2000 to 2019.

Each French firm trading outside the EU files a compulsory customs form whenever its merchandise value is greater than €1,000 (or 1,000 kilos).[6] The customs database specifies the month and year of filing, export or import flow, the partner country, product code (CN-code), time-invariant French firm identifier (SIREN), weight or volume transacted, and value. After 2011, the invoice currency is available.

I link customs information with two data sets containing firm characteristics. For the period 2000 through 2007, I use the FICUS data set (*Fichier Complet Unifié de Suse*). For the period 2008 through 2019, I use the FARE data set (*Fichier Approché des Résultats d'Esane*). FICUS and FARE are administered consecutively (FICUS through 2007, FARE from 2008), but the balance-sheet and income-statement variables this paper uses are defined identically in both. The 2007/2008 handover therefore introduces no break in any series I rely on. These data sets contain balance sheets and income statements from administrative tax records integrated with information on employment, firm age, and other business characteristics gathered by INSEE (*Institut National de la Statistique et des Études Économiques*). The sample covers the universe of corporations and medium-sized "non-commercial" firms active in France. I integrate administrative information with the extent of their foreign activity using the datasets LIFI (*Liaisons Financières entre Sociétés*), which identifies the ownership links between enterprises operating in France, and OFATS (*Outward Foreign Affiliates Statistics*), a survey containing the structure and activity of foreign affiliates of French firms.

4.2 Sample and invoicing patterns

The paper focuses on a sample of 19,115 firms. Together they account for 61.5 percent of extra-EU exports and 60.0 percent of imports. This sample is a deliberate subset. The customs universe covers 351,042 firms, and Table C.1 compares the full trade sample with this paper's sample. The restriction is for identification, not data availability. Shift-share identification needs each firm to be hit by the exchange-rate shock repeatedly, over enough years, for its exposure to be precisely estimated (Borusyak et al. 2022). I keep firms that trade outside the EU

[6] The threshold was discontinued in 2010, and all the results for the period 2011 through 2018 represent virtually the totality of extra-EU trade. Whenever I extend the sample to the period 2000 through 2018, I homogenize the data to reflect the pre-2010 threshold. For more details see Table C.1 and Bergounhon et al. (2018).

in every year of the window (2000–2018), so a firm's invoicing exposure is a stable, repeatedly shocked object. Results are virtually unchanged on an unbalanced sample (Figure B.1).

Table 1: Firm-level descriptive statistics

	Net exporters		Net importers	
	Top 1%	Others	Top 1%	Others
Firms	118	11,663	74	7,260
Panel A. Export flow				
Contribution to total exports (%)	52.8	38.8	4.0	4.4
Invoice currencies	10.0	2.0	3.6	1.7
HS6 products	269.2	44.1	161.8	32.7
Partner countries	82.0	20.3	38.7	13.2
Currencies per country	1.57	1.10	1.31	1.09
Currencies per country-HS6	1.17	1.03	1.10	1.02
EUR-invoiced share (%)	51.9	92.9	61.0	91.6
USD-invoiced share (%)	37.3	5.2	30.8	6.7
Invoicing never switches (%)	97.7	99.2	97.8	99.3
Panel B. Import flow				
Contribution to total imports (%)	21.8	9.1	33.8	35.3
Invoice currencies	13.2	3.2	8.1	3.2
HS6 products	261.0	28.7	250.9	56.9
Partner countries	37.4	7.2	27.7	8.4
Currencies per country	1.88	1.34	1.72	1.47
Currencies per country-HS6	1.18	1.08	1.21	1.13
EUR-invoiced share (%)	39.6	53.5	52.9	45.4
USD-invoiced share (%)	48.1	36.4	43.9	47.7
Invoicing never switches (%)	96.3	97.8	97.4	97.9

Notes: The 19,115 firms in the sample are split in net exporters and net importers. Within each group the table separates between the top 1% by gross trade and the rest. Panel A describes each group's export flow and Panel B its import flow, over 2011–2018, the period when invoice currency is observed. *Contribution* is the group's share of total sample exports (imports). Count rows are unweighted cross-firm means. Currencies, HS6 products, and partner countries are distinct counts per firm. The per-country and per-country-HS6 rows average the within-combination currency count across a firm's destinations. EUR and USD shares are firm-level invoiced value shares. *Invoicing never switches* is the share of the group's single-currency products (firm $\times$ HS6 $\times$ partner $\times$ shipment terms) whose invoicing regime is unchanged in every year it is traded.

Table 1 describes the trade characteristics of the sample, split into net exporters and net importers. A firm is a net importer when its imports exceed its exports in the overall sample. Within each group, I distinguish between the top 1% of firms by gross trade versus the rest to give a sense of the granularity of the trade data. 38.4 percent of firms are net importers. The typical firm trades in many products and destinations, a median of 38 HS6 products across 15 partner countries, using 3 invoice currencies. Activity is heavily concentrated within each group, with the top 1% of net exporters accounting for over half of exports and the top 1% of net importers for about a third of imports. Net exporters are also generally importers. Three features of the data are worth stating.

Trade is invoiced almost entirely in euros and dollars, and imports are more dollar-invoiced than exports. Dollar invoicing accounts for 40.5% of exports and 51.1% of imports (euro invoicing for 50.5% and 43.6%). Only 30.7% of dollar-invoiced trade is with the United States. Imports are more dollarized than exports, and the largest net exporters are more dollarized than the rest. This heterogeneity in dollarization matters later for the incidence of exchange-rate shocks on aggregate outcomes.

Invoicing is stable over time. Currency is a near-permanent attribute of a trade relationship. 98.8% of single-currency products never switch invoicing regime over the sample, and the "invoicing never switches" row of Table 1 shows this near-permanence holds within every group. Product-level invoicing is near-absorbing, with a single-currency product keeping its regime from one year to the next with probability 95.6–98.2%, depending on the regime (Table C.2).

Invoicing variation is across product-country-firm, not within them. The dollar-invoicing share varies sharply across partner-country and industry combinations (Figure B.2). The median firm invoices in 3 currencies overall, but that multiplicity is entirely across its products and destinations. At the finest level, the firm $\times$ HS6 $\times$ partner combination, the average number of invoice currencies is barely above one (1.02–1.21 across the groups of Table 1, the "currencies per country-HS6" row).

5 Exchange Rates and Trade Flows

5.1 Measuring invoice-currency exposure

In this section, the unit of analysis is a *trade relationship* $c = (f, g, n, d)$, consisting of a firm f, an HS6 product g, a partner country n, and a flow direction $d \in \{\text{export}, \text{import}\}$. I refer to this firm–product–country combination (in a given direction) as a trade relationship throughout. Some appendix exercises split a relationship further, by invoice regime or by shipment terms, into what their table notes call a product; each note states the exact unit. Invoice currency is recorded only from 2011 on. To carry exposure back to 2000 I assume that, within a relationship, the invoicing currency is a near-permanent attribute.[7] For each relationship and invoicing currency $j \in \{€, *, \$\}$ (euro, the partner currency $*$, and the dollar), I compute the

[7] This assumption is common in the literature and supported by section 4.2. Amiti et al. (2022) make the same assumption on Belgian data, extrapolating firm-destination invoicing shares backward to years they do not observe on the strength of the high persistence of currency choice.

value share of currency j over the years in which invoicing is observed,

$$\omega_c^j = \frac{\sum_{t\geq 2011} V_{ct}^j}{\sum_{t\geq 2011} V_{ct}}, \qquad \sum_j \omega_c^j = 1, \tag{32}$$

where V_{ct}^j is the euro value of relationship c invoiced in currency j in year t, and I assign ω_c^j to relationship c in every year of the 2000–2018 sample. Relationships that trade before 2011 but are never observed after it carry no imputed share and drop from the baseline. The leave-one-out construction of Table C.4 assigns them their product–partner cell's currency mix net of the firm's own trade and delivers the same value pass-through.

This carry back can at most cause attenuation bias but it is in fact a strong predictor of past invoicing regimes for two reasons. First, invoicing is a near-permanent attribute of a relationship (Tables 1 and C.2). Second, a relationship prices a given product to a given destination in essentially one currency (Table 1), so the imputed object is most often a stable 0/1 exposure.

The robustness section in appendix confirms the stability of the results regardless of the time sample or the construction of invoice share index. Re-estimating on the 2011–2018 sample alone, where invoicing is observed and nothing is imputed, leaves the results unchanged (Table C.3). Table C.4 shows the value pass-through is the same whether the share is the pooled 2011–2018 average, the 2011 value alone (which uses less look-ahead information), or a leave-one-out cell average.

5.2 Exchange-rate pass-through: prices, volumes, values

Before measuring the profit response of exposed firms, this section establishes how prices, quantities, and values behave at the border, at the level of a single trade relationship.

To that end I estimate a local projection that interacts the invoicing shares of (32) with the exchange-rate movement. The unit of observation is a trade relationship c followed quarterly between 2000 and 2018. Exports and imports are estimated separately. The invoicing shares ω_c^j weight three regimes. Euro- and partner-invoiced flows are interacted with the euro/partner log change in value $\Delta e_t^{€/*}$, interpreted as a euro devaluation against the partner currency. Dominant (dollar)-invoiced flows are priced in a currency that is foreign to both France and the partner. In this case I split the exposure into a euro/dollar leg $\Delta e_t^{€/\$}$ and a dollar/partner leg $\Delta e_t^{*/\$}$, the movement in the buyer's partner-currency price of a dominant currency invoice. These are the two movements that the seller and the buyer perceive under dominant-currency pricing.

Let y_{ct} denote the log unit value (in euro terms), the log real volume, or the log value (in euro terms) of relationship c in quarter t. The local projection is defined as,

$$\begin{aligned} y_{c,t+h} - y_{c,t-1} = \sum_l \Big[& \underbrace{\beta^{€,y}_{h,l}\,\omega^{€}_c\,\Delta e^{€/*}_{t-l}}_{\text{Euro}} + \underbrace{\beta^{*,y}_{h,l}\,\omega^{*}_c\,\Delta e^{€/*}_{t-l}}_{\text{Partner}} \\ & + \underbrace{\beta^{\$,y}_{h,l}\,\omega^{\$}_c\,\Delta e^{€/\$}_{t-l}}_{\text{Dominant}} + \underbrace{\gamma_{h,l}\,\omega^{\$}_c\,\Delta e^{*/\$}_{t-l}}_{\text{Buyer leg}} \Big] \\ & + \Gamma'_h X_{ct} + a_c + d_t + \varepsilon^h_{ct}, \end{aligned} \tag{33}$$

with l running over the contemporaneous quarter and its lags up to one year. The vector X_{ct} collects trade-weighted partner-country CPI and nominal GDP, contemporaneous and lagged. The terms a_c and d_t are relationship and quarter fixed effects. The pre-shock coefficients test for anticipation effects and pre-trends. Fixed effects and controls enter only to absorb demand or supply shocks that might correlate with the exchange rate.[8] The impulse response plotted in Figure 1 is the sequence of contemporaneous coefficients $\beta^j_{h,0}$ traced over h, from three quarters before the shock to eight quarters after it.[9]

The coefficient reads as a pass-through to a euro depreciation. The three regimes deliver distinct predictions. Price stickiness implies that a relationship's exposure depends on the currency in which goods are priced. When expressed in euro terms, euro-invoiced flows should not respond to the euro/partner exchange rate, $\beta^{€,p}_{h>1,0} \simeq 0$, while partner-invoiced and dollar-invoiced flows should respond close to one-to-one, $\beta^{*,p}_{h>1,0} \simeq \beta^{\$,p}_{h>1,0} \simeq 1$.

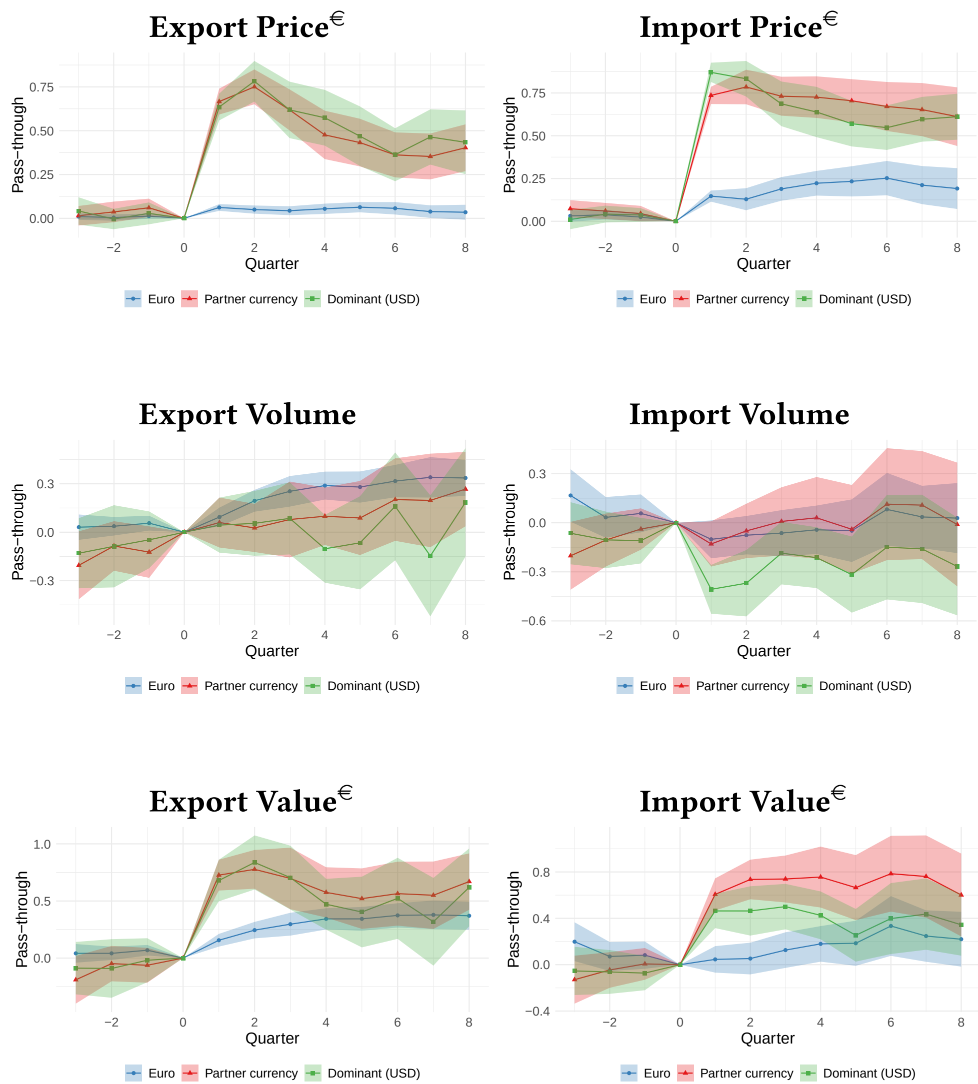
The premise holds. A euro depreciation passes into the euro unit values of foreign-invoiced flows close to one-for-one (around 0.8 in the short term) but barely moves euro-invoiced ones. Volumes respond, but by less than prices. The relevant price that demand should respond to is the one the buyer faces. That is the euro price for the French importer and the partner-currency price for the foreign buyer of a French export. Measured this way, the implied trade elasticity conditional on the exchange-rate shock is about 0.3 to 0.4 in absolute value, similar across invoicing regimes and on both sides of trade.[10] Because it is below one, the quantity

[8]Such shocks are rare at this frequency, a reflection of the well-known disconnect between exchange rates and macroeconomic fundamentals. The annual fixed-effect ladder in Table C.5 confirms it. The dominant value pass-through stays positive and significant as the fixed effects are tightened, up to a separate exchange-rate slope for every partner country, which is what one expects when the shock is uncorrelated with unobservables.

[9]For dollar-invoiced flows I include both legs of the bilateral movement, the euro/dollar leg and the dollar/partner leg $\Delta e^{*/\$}_t$. The dominant coefficient plotted is the one on the euro/dollar leg, $\beta^{\$}_{h,0}$, which reads as the response to a uniform euro depreciation against all currencies and is the best identified, since it exploits a currency the two trading partners do not share (section 6.2).

[10]Conditional elasticities are the ratio of the quantity coefficient to the buyer-currency price change in Table C.6. For dollar-invoiced exports the relevant leg is the dollar/partner one, where the sticky-dollar price moves the buyer's partner-currency price almost one-for-one. The dollar leg gives $-0.27/0.76 \approx -0.36$ for imports and $-0.29/0.90 \approx -0.32$ for exports. Partner-invoiced exports give $0.17/(-0.43) \approx -0.39$ and euro-invoiced exports $0.32/(-0.98) \approx -0.32$. Partner-invoiced imports show no significant volume response. Euro-invoiced

Figure 1: Exchange-rate pass-through by invoicing regime, quarterly impulse responses



Notes: Local-projection impulse responses, (33), of the log euro unit value (price), log real volume, and log euro value of a trade relationship to a euro depreciation, by invoicing regime (euro, partner currency, and dominant dollar). Export flows are to the left and imports to the right. Each point is the contemporaneous coefficient $\beta^{j}_{h,0}$ at horizon h, from three quarters before the shock to eight after. For dollar-invoiced flows the plotted dominant coefficient is the euro/dollar leg. All panels include relationship and quarter fixed effects, with trade-weighted partner-country CPI and nominal GDP as controls. Standard errors are double-clustered by relationship and quarter. Shaded areas are 95% confidence intervals.

response attenuates the revaluation without overturning it. The value response shows it directly. Values inherit most of the price response, and foreign-priced trade behaves in the short run like a position denominated in foreign currency. The differential value effect across invoicing regimes is statistically distinguishable for roughly the first five to six quarters after the

imports give $-0.22/0.14$, but the euro price barely moves and the value response is zero, so the ratio is imprecise and these flows carry no valuation exposure in any case.

shock and fades thereafter as inherited prices reset and quantities become more responsive, which delimits the short-run window this paper studies.[11]

I deliberately do not estimate heterogeneous effects by firm size, market share, or import and export intensity, nor do I condition on them. These characteristics shape both how trade is invoiced and how it responds to exchange rates (Amiti et al. 2022, Crowley et al. 2020), so conditioning on them would restrict identification to the *unexplained* variation in currency choice and redefine the estimand, as in Amiti et al. (2022).[12] The object of interest is instead the unconditional pass-through, the full ex-post effect of being invoiced in a foreign currency. Part of this differential may itself reflect selection. Firms and flows that price in a foreign currency need not be comparable to those that price in euros. That is by design. The estimand is the average revaluation carried by flows as they are actually invoiced in the data, not the effect of randomly reassigning a flow's currency. I treat selection into invoicing as part of the estimand rather than as a bias to remove. Identification only requires exogeneity of the exchange-rate shock, not of the currency choice. Even so, the appendix shows that adding the full Amiti et al. (2022) control set leaves the results virtually unchanged (Table C.8).

6 Firm-Level Evidence

6.1 Measuring exposure at the firm level

To first order a firm's exchange-rate profit response is proportional to the value of its net preset foreign-currency positions, times the exchange-rate movement, as in (20). The first way to measure preset positions follows the effective-exchange-rate literature and considers a firm's net trade by partner's currency. Let $V_{c,t-1}$ be the firm's lagged trade in relationship $c = (f, g, n, d)$ expressed in euros, for firm f, HS6-product g, partner country n, trade flow d. The net *trade* position of each firm f against partner n is exports (revenue) minus imports (cost),

$$N^{T,n}_{f,t-1} = \sum_{c \in f:\, n(c)=n,\, d=\text{exp}} V_{c,t-1} - \sum_{c \in f:\, n(c)=n,\, d=\text{imp}} V_{c,t-1}. \tag{34}$$

[11] The annual long panel (Table C.6) confirms these responses at the annual frequency. They are the same responses visible, more noisily, in the quarterly impulse responses. A further check, Table C.7, re-runs the benchmark specification with relationship entry and exit as the outcome, in place of prices, volumes, and values. The exchange-rate movement does not pull dollar-invoiced relationships into or out of trade differentially.

[12] By the logic of Engel (2006) and Gopinath et al. (2010), a flow is invoiced in the currency that minimizes the covariance between its desired price and the exchange rate, so the chosen currency is a sufficient statistic for the flow's desired pass-through. It already encodes the cost structure, markups, and competitor pricing that generate it. This is also why conditioning on the determinants is neither necessary nor clean in this context. Currency choice is a complex process driven by unobserved financial, institutional, and informal factors as much as by economic forces, and it is beyond this paper's purpose to observe all such determinants.

The second measure (and main focus of this paper) is the net *invoice currency* j position:

$$N^{I,j}_{f,t-1} = \sum_{c \in f:\, d=\text{exp}} \omega^j_c \, V_{c,t-1} - \sum_{c \in f:\, d=\text{imp}} \omega^j_c \, V_{c,t-1}. \tag{35}$$

$N^{I,j}_{f,t-1}$ weights each pre-set value of a trade relationship by the share of value invoiced in currency j, defined as in (32).

I then define a *trade-weighted index* that revalues every partner position at that partner's bilateral euro rate,

$$T_{ft} = \sum_n N^{T,n}_{f,t-1} \, \Delta e_t^{€/*_n}, \tag{36}$$

where $*_n$ denotes the currency of partner n, and the *invoice-weighted index*, the empirical counterpart of the model's FX profit term, that revalues each currency bucket against the euro-to-invoice currency rate,

$$I_{ft} = \sum_j N^{I,j}_{f,t-1} \, \Delta e_t^{€/j}, \tag{37}$$

with $\Delta e_t^{€/€} = 0$, so that euro-invoiced positions carry no valuation term and I_{ft} loads only on foreign-currency fluctuations related to invoicing shares. For $j = *$ the index revalues each partner-invoiced position at its own partner currency's euro rate, as the channel split in section 6.2 makes explicit.

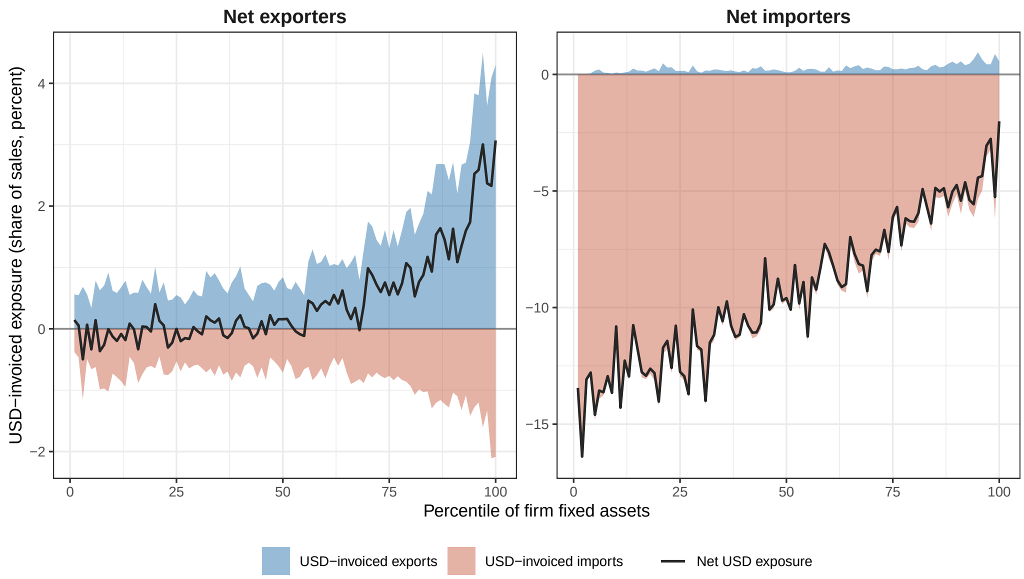
The trade-weighted index T_{ft} is the object a standard sticky price model would predict to move cash flow. Under producer-currency pricing (PCP) or local-currency pricing (LCP) the only relevant exchange rate leg is the euro/partner rate, as (21) shows. In addition, if conditional trade elasticities were above one, cash flow should co-move positively with T_{ft}. A euro depreciation would raise the cash flow of net exporters and lower that of net importers, in proportion to their net trade positions.[13]

Figure 2 plots the dollar net invoice position $N^{I,\$}_{f,t-1}$, the $j = \$$ case of (35), and measures how intensely firms are exposed. It ranks firm-years into percentiles of fixed assets and shows

[13]With high (above one) trade elasticities, under PCP a euro depreciation lowers the buyer-currency price of French goods one-for-one and export volumes rise more than proportionally. Imported inputs invoiced in the partner currency become more expensive in euros. The expenditure-switching gain $\sigma_i \Pi_{i,t-1}$ in (21) then outweighs the input-cost revaluation, so cash flow rises with the net trade position. Under LCP quantities are insulated on impact, because prices are preset in the buyer's currency. The euro value of foreign-currency export receipts still rises, whatever the elasticity. With elasticities below one, as measured in section 5.2, the PCP prediction reverses. The expenditure-switching gain is too small to offset the input-cost revaluation. A depreciation can then lower the cash flow even of a net exporter, and the response tracks gross imported inputs rather than the net trade position. The euro-partner currency exchange rate is still the relevant leg to measure this effect. The LCP prediction survives low elasticities, but it is itself a valuation effect, and it also ties the revaluation to the euro/partner leg.

mean USD-invoiced exports and imports and the resulting net position, all relative to sales. The dollar case is the relevant one. Euro positions carry no valuation term and other foreign currencies invoice a marginal share of French trade (section 4.2). The object of interest in the figure is exposure intensity, which governs the magnitude of the shock relative to the firm's operations.

Figure 2: Net USD-invoiced exposure by firm size and net-trade position



Notes: Firm-years are ranked into percentiles of fixed assets within each group, net exporters left and net importers right. Shaded areas are mean USD-invoiced exports (positive) and imports (negative) over sales. The line is the mean net exposure.

Intensity is asymmetric across net exporters and net importers. Net importers are short the dollar in every asset bin, and most intensely at the bottom of the size distribution. Their net position runs to about 14 percent of sales in the lowest bins and declines to 2–5 percent at the top. Net exporters are less exposed for two reasons. Their dollar operations are relatively smaller to begin with. Gross USD flows average about 2 percent of sales against roughly 10 percent for net importers.[14] The exposure they carry is also operationally hedged. Dollar exports roughly match dollar imports, so the net position sits near zero up to the 55th percentile.[15]

Table C.9 in the appendix provides additional details about the two groups of firms and their positions. Net importers and net exporters are comparable in size. However, net importers carry a heavier debt service. The smallest net importers pay 2.3 percent of capital in interest, against 1.4 percent for net exporters. Net importers' main external liability is trade

[14]This first margin is hidden in Table 1, which scales by gross trade rather than operations.

[15]The asymmetry is visible only when the two populations are kept apart, since pooling them masks it (Figure B.3).

credit from suppliers, larger than bank and bond debt at every size bin. Cash holdings relative to capital fall with firm size for both net exporters and importers. This is part of the justification for using size as a proxy for the marginal value of internal funds (Figure B.4 reports the full correlate set). Table C.9 in the appendix also highlights how the most intense exposure and the heaviest debt service sit with the same small net importers group.

6.2 Invoice-currency exposure and cash flow

To compare the trade-weighted and invoice-weighted index I estimate them jointly,

$$\frac{y_{ft}}{A_{f,t-1}} = \beta^T \frac{T_{ft}}{A_{f,t-1}} + \beta^I \frac{I_{ft}}{A_{f,t-1}} + \mu' X_{ft} + a_f + d_{s(f),t} + u_{ft}. \tag{38}$$

The dependent variable is a firm flow scaled by lagged total assets $A_{f,t-1}$, following standard practice in corporate finance (e.g., Hubbard 1998, Rauh 2006, Lewellen and Lewellen 2016). The coefficients β read in euros of y flow per euro of exposure.

The invoice-weighted index admits a finer specification that splits it into currency-specific channels. This replaces I_{ft} by its partner-currency and dominant-currency components,

$$\frac{y_{ft}}{A_{f,t-1}} = \underbrace{\beta^{€} \frac{I^{€}_{ft}}{A_{f,t-1}}}_{\text{Euro}} + \underbrace{\beta^{*} \frac{I^{*}_{ft}}{A_{f,t-1}}}_{\text{Partner}} + \underbrace{\beta^{\$} \frac{I^{\$}_{ft}}{A_{f,t-1}}}_{\text{Dominant}} + \mu' X_{ft} + a_f + d_{s(f),t} + u_{ft}, \tag{39}$$

where the dominant channel revalues the net dollar position at the euro/dollar rate, $I^{\$}_{ft} = N^{I,\$}_{f,t-1}\,\Delta e^{€/\$}_t$ when the US is not a partner, and the partner channel sums each partner-invoiced net position over its partner-currency valuations, $I^{*}_{ft} = \sum_{*} N^{I,*}_{f,t-1}\,\Delta e^{€/*}_t$. Together the channels reproduce the aggregate index (37). Dollar trade with the United States, where the dollar is the partner currency, enters the partner channel. The euro channel is a placebo, $I^{€}_{ft} = \sum_{*} N^{I,€,*}_{f,t-1}\,\Delta e^{€/*}_t$, where $N^{I,€,*}_{f,t-1}$ is the euro-invoiced net position with partners whose currency is $*$, multiplied by the euro–partner rate. It provides a parallel index to the relationship-specific euro invoice response in section 5.2.

X_{ft} collects partner-country macroeconomic controls (trade-weighted CPI and nominal GDP), a_f is a firm fixed effect, and $d_{s(f),t}$ an industry$\times$year fixed effect. Adding nonparametric country-of-exposure $\times$ year slopes leaves the coefficients virtually unchanged, though less precisely estimated in more disaggregated specifications (Table C.10). For dominant-invoiced flows the specification also includes the dollar/partner leg $I^{\$,*}_{ft} = N^{I,\$}_{f,t-1}\,\Delta e^{*/\$}_t$, the change in the price the partner faces in its own currency. The leg is separately identifiable only under dominant currency pricing, where invoicing and partner currency diverge, and its inclusion lets $\beta^{\$}$ isolate the valuation effect. $\beta^{\$}$ is therefore the preferred measure beyond this section.

The indices I^j_{ft} are shift-share objects, with predetermined net positions as shares and the exchange-rate movement as the shock. The shares are not random. Net dollar exposure rises with size, trade intensity, sector, and net-importer and multinational status (Figures 2 and B.4), so on their own the shares cannot identify the valuation effect of foreign currency operations (Goldsmith-Pinkham et al. 2020). Identification rests instead on the exchange rate shock, under two conditions made precise in Appendix A. First, the euro-dollar movement is quasi-random with respect to the exposure-weighted residual of the most exposed firms. Second, identification leans on many serially uncorrelated annual shocks rather than one episode, and the 2001–2019 window supplies one of the longest such series in the literature. Consistent with the first condition, the valuation index is uncorrelated with firms' lagged residualized outcomes (Figure B.5). Consistent with the second, no single year drives the estimate (Appendix A). This is due to exchange rates behaving like hard-to-predict random walks largely disconnected from fundamentals. Moreover, France sits inside a currency union whose monetary policy responds little to French firms. A final check fixes the shares outright, holding each firm's net positions at the 2000 base period so that exposure varies only through the exchange rate. The estimates show the same pattern, and the panel, which no longer needs rolling customs positions, extends through 2021 (Table C.11).

Table 2 includes the results from specifications in (38)-(39). The outcome is the firm's gross operating surplus, an EBITDA analogue.[16] Estimated alone, the trade-weighted index yields 13 cents of cash flow per euro of exposure (column 1). Adding the invoice index drops the coefficient to 6 cents against an invoice coefficient of 41 cents (column 2). That I absorbs T is direct evidence the cash-flow variation is predominantly an invoice valuation effect. This can account for the small, unstable exposures reported by studies built on trade-weighted rates (Jorion 1990, Bartram et al. 2010, Alfaro et al. 2023a).

Column 3 splits the invoice index by currency. The euro channel is insignificant, the null the model predicts for euro-priced flows. The partner channel is positive, and the dominant channel reaches 0.67 per euro of exposure, three to five times the trade-weighted estimate. This mirrors the border result of section 5.2. Columns (4)–(6) further decompose the dominant channel for a sharper identification.[17] Column 4 recovers the sign the accounting requires, a dollar appreciation raising cash flow through dollar exports and lowering it through dollar imports, with the import flow larger and precisely estimated (-0.73) and the export flow positive but imprecise. Re-estimated on net exporters and importers separately (columns 5–6), the

[16] Gross operating surplus is formed above the financial result. Dollar-denominated debt, treasury and intragroup balances, and the settlement of currency derivatives are recorded below it. The discussion after Table 2 reads the financial block directly (Table C.12).

[17] Again, this coefficient loads on the euro/dollar rate, a currency external to both France and the partner, so its variation is plausibly orthogonal to any single trade relationship. Restricting to dollar-priced trade with non-US partners lets the specification absorb partner-currency movements and firm-by-partner trends.

Table 2: Cash-flow responses to trade- and invoice-weighted exchange-rate exposure

	Cash Flow					
	Pooled sample				Net exporters	Net importers
	(1)	(2)	(3)	(4)	(5)	(6)
Trade exposure, T_{ft}	0.1335***	0.0582*				
	(0.0339)	(0.0323)				
Invoice exposure, I_{ft}		0.4119***				
		(0.0757)				
Euro exposure, $I_{ft}^{€}$			0.0288			
			(0.0266)			
Partner exposure, I_{ft}^{*}			0.3757***			
			(0.0939)			
Dominant exposure, $I_{ft}^{\$}$			0.6672***			
			(0.1601)			
Dominant exposure, $I_{ft}^{\$,\text{exp}}$				1.285	1.270	2.082
				(0.8764)	(0.7923)	(1.414)
Dominant exposure, $I_{ft}^{\$,\text{imp}}$				-0.7281***	-0.5950**	-0.5420***
				(0.1842)	(0.2261)	(0.1548)
Observations	202,941	202,941	202,941	202,941	121,607	81,334
R^2	0.63477	0.63527	0.63516	0.63517	0.64961	0.66444
Within R^2	0.19446	0.19556	0.19531	0.19532	0.17244	0.19957
Firm fixed effects	✓	✓	✓	✓	✓	✓
Industry-Year fixed effects	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is cash flow scaled by lagged total assets. The coefficients read as euros of cash flow per euro of exposure. Column (1) includes trade-weighted exposure alone. Column (2) adds invoice-weighted exposure. Column (3) splits the invoice index into euro, partner-currency, and dominant (dollar) net channels. Column (4) splits the dominant channel into its export and import flows, $I_{ft}^{\$} = I_{ft}^{\$,\text{exp}} - I_{ft}^{\$,\text{imp}}$, each term revaluing the gross dollar-invoiced flow at the euro/dollar rate. Columns (1)–(4) use the pooled sample. Columns (5)–(6) re-estimate the dominant flows on the net-exporter and net-importer subsamples. All columns include partner-country controls (trade-weighted CPI and nominal GDP) and firm and industry×year fixed effects. Columns (3) to (6) include the dominant exposure loaded on the dollar-partner exchange rate, $I_{ft}^{\$,*}$. Standard errors in parentheses, double-clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

import flow stays negative and significant for both groups. This isolates a general valuation effect rather than an artifact of a specific firm group or exposure kind.

One potential objection to the estimates in Table 2 is that dollar invoicing may proxy for other dollar exposures. A firm that prices its trade in dollars may also borrow in dollars, and one euro-dollar movement revalues the trade and the debt at once. Two facts lower this concern. First, Table C.12 regresses financial income, the accounting variable where revaluations of financial assets are recorded, on the same dominant-currency exposure. The response is indistinguishable from zero for private, small, and medium firms, and turns mildly negative and significant only for public (-0.09) and large (-0.05) firms, the groups best able to carry financial hedges. A negative financial-income response to an appreciation that raises the same firms' operating surplus is the sign a hedge leaves when it offsets an operating position but it remains a small fraction of the cash-flow coefficient across all firms. Dollar funding in France is scarce to begin with, about two percent of bank claims and liabilities in the BIS locational statistics, and concentrated in very large firms (Berthou et al. 2022). Second, the two trade flows respond with opposite signs, -0.73 on dollar imports and positive on dollar exports. A dollar-credit tightening would instead lower cash flow through both flows at once.[18] In short, the design cannot purge an unobserved firm-level dollar-financing position, and does not claim to.[19] However, the results shows that the channel it identifies behaves with the sign pattern and balance-sheet allocation of an operational valuation effect.

6.3 Sources and uses of the cash-flow shock

By the firm's sources-and-uses identity, every euro of cash flow is accounted for by a change in a balance-sheet or distribution item (Lewellen and Lewellen 2016),

$$\underbrace{\Delta\text{Cash}_{ft} + \Delta\text{NWC}_{ft} + \text{CapEx}_{ft} + \text{Div}_{ft}}_{\text{uses}} - \underbrace{\left(\Delta\text{Debt}_{ft} + \text{Issues}_{ft} + \text{FinInc}_{ft}\right)}_{\text{external finance and financial income}} = \text{CF}_{ft}. \quad (40)$$

Here ΔCash_{ft} is the change in cash and equivalents, ΔNWC_{ft} the change in net working capital, CapEx_{ft} capital expenditure, and Div_{ft} dividends paid. On the financing side, ΔDebt_{ft} is net debt issuance, Issues_{ft} net equity issuance, and FinInc_{ft} financial income. CF_{ft} is operating cash flow. I scale each term by lagged assets, as in (39), and regress it on the same dominant-currency exposure with the same controls and fixed effects.[20] Because the dependent variables approximately satisfy (40) term by term and the regressors are common across the equations, the component coefficients reconcile to the total cash-flow coefficient up to a residual.[21] Figure 3 stacks them into a waterfall chart.

On the sample where every component of the identity is observed, the total cash-flow re-

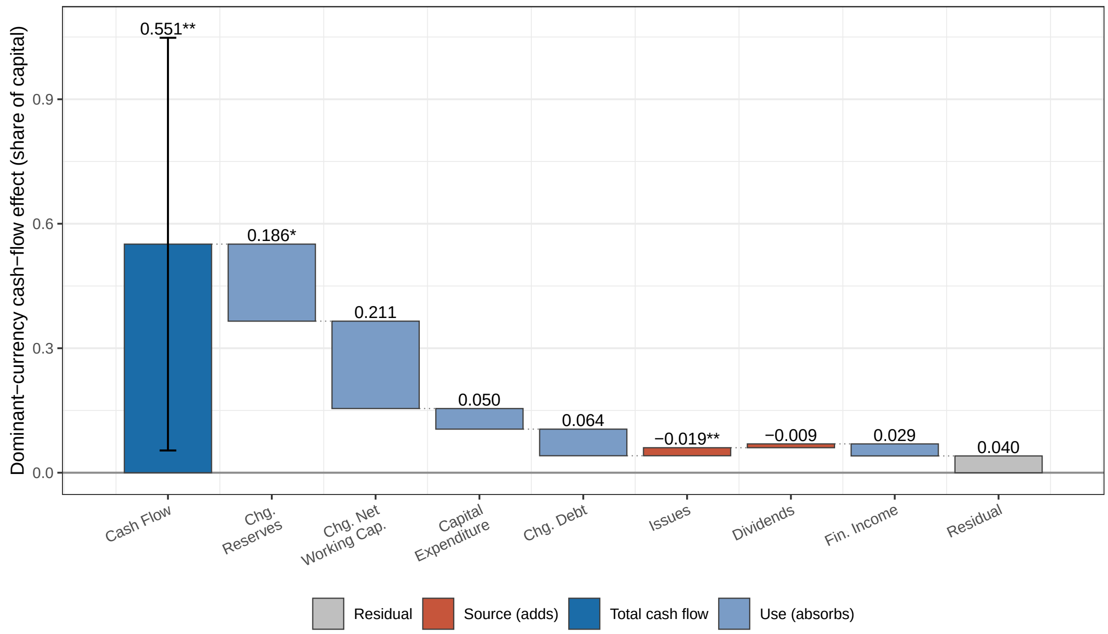
[18] If instead the correlated dollar financing came with a firm's net trade position, the negative import response would be confined to the net importers. It stays negative and significant in both subsamples (columns 5–6).

[19] Appendix D.4 shows that when invoicing regimes differ in exchange-rate-sensitive items beyond the posted price, the regression recovers the total exposure to each leg and the sticky-price invoice exposure wedge is only one component of it.

[20] The design avoids the accounting-identity critique of investment–cash-flow regressions (Sánchez-Vidal 2023). The regressor is an exchange-rate valuation shock external to (40), not one identity component projected on another, and estimating the full set of components keeps any single coefficient from absorbing the omitted terms.

[21] Component measures built from balance-sheet differences still carry the measurement error Lewellen and Lewellen (2016) emphasize, from acquisitions, write-offs, and currency translation, which the waterfall residual absorbs.

Figure 3: Sources and uses of the dominant-currency cash-flow response



Notes: Each bar is a regression coefficient on dominant-currency exposure. The leftmost bar is the total cash-flow response with its 95% confidence interval. Each subsequent bar is the share of that total absorbed (a use) or supplied (a source) by a balance-sheet component of the sources-and-uses identity (40) (Lewellen and Lewellen 2016). The grey residual closes the decomposition. Each component is scaled by lagged total assets and estimated on the sample in which all components are observed.

sponse to dominant-currency exposure is 0.55, close to the 0.67 benchmark of Table 3. The gap reflects the smaller common sample, not the size of the effect. Most of the response accumulates as liquidity. The change in cash reserves absorbs 0.19 (10% significance), a third of the total, and net working capital a comparable but imprecise 0.21. Together, the two liquid buffers take about 70% of the shock. The remaining uses are flat. Capital expenditure (0.05), debt (0.06), dividends (-0.01), and equity issues (-0.02) each absorb little, so there is neither a real-spending nor a payout response on average.

6.4 Average real effects

Table 3 reports the average cash flow and real effects on salaries, employment, and investments. Columns 1–3 put every outcome over the same lagged-assets denominator, so coefficients read euro-for-euro. Columns 4–7 put each outcome in its own natural units, a revenue margin, growth rates, and a capital rate.

Dominant-currency exposure raises cash flow by 0.67 and salaries by 0.23, while total capital expenditure does not move (0.03, columns 1–3).[22] In each outcome's natural units

[22]French business accounts (FARE/FICUS, drawn from the tax *liasse fiscale*) report no cash-flow-statement

Table 3: Invoice-currency exposure and firm outcomes

	Dollar-for-dollar			Natural units			
	Cash Flow	Salaries	Tot. CAPEX	CF Margin	Empl. Growth	Salary Growth	Investment Rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Euro exposure, $I^{€}_{ft}$	0.0288	-0.0089	-0.0220	0.0461**	0.0265	0.0680	0.0525
	(0.0266)	(0.0165)	(0.0137)	(0.0170)	(0.0643)	(0.0692)	(0.0892)
Partner exposure, I^{*}_{ft}	0.3757***	0.1171**	-0.0364	0.2281**	-0.3704*	0.1433	0.4412
	(0.0939)	(0.0466)	(0.0769)	(0.0909)	(0.1969)	(0.1152)	(0.3492)
Dominant exposure, $I^{\$}_{ft}$	0.6672***	0.2303**	0.0306	0.3178***	0.1631	0.6731***	0.3030
	(0.1601)	(0.0798)	(0.0320)	(0.0857)	(0.1622)	(0.2277)	(0.2587)
Observations	202,941	202,941	202,937	219,539	214,182	218,621	185,506
R^2	0.63516	0.87327	0.18889	0.63428	0.24483	0.17229	0.17485
Within R^2	0.19531	0.35339	0.00293	0.15451	0.14626	0.03047	0.00373
Firm fixed effects	✓	✓	✓	✓	✓	✓	✓
Industry-Year fixed effects	✓	✓	✓	✓	✓	✓	✓

Notes: Columns (1)–(3) report dominant (dollar), euro, and partner-currency exposure on cash flow, salaries, and total CAPEX, each scaled by lagged total assets. Columns (4)–(7) report the same three currency channels with each outcome in its natural units, the cash-flow margin over sales, employment and salary growth in log changes, and the investment rate over capital. Total CAPEX is the year-over-year change in gross fixed assets. The investment rate uses the reported gross investment flow over a perpetual-inventory capital stock with 10% depreciation. All controls are as in column (3) of Table 2. Standard errors in parentheses, double-clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(columns 4–7) the cash-flow margin over sales increases 0.32% and salary growth 0.67% per euro of invoice-valuation exposure over lagged assets, while employment growth and the investment rate over capital are positive but insignificant. Salary growth with flat employment growth pins the margin. The compensation bill rises almost entirely through compensation per worker, not headcount, since $\Delta \log \text{comp. bill} = \Delta \log \text{emp} + \Delta \log \text{comp./worker}$ holds exactly. The accounting compensation-bill line bundles social contributions, bonuses, and benefits, so base pay cannot be separated from bonuses or benefits. The pattern across invoice currencies is consistent as well, with large effects only for partner- and dollar-invoiced exposures.[23]

capital-expenditure line of the kind standard in corporate finance. Firms instead file a fixed-asset schedule (*tableau des immobilisations*) that records, for each asset class, the gross book value and its within-year movements. These are acquisitions and in-house creations, contributions (*apports*), reclassifications, disposals, and revaluations. The two capital-spending measures here are the closest analogues built from these entries. Total CAPEX (column 3) is the year-on-year change in gross fixed assets. The investment rate (column 7) instead uses the directly reported gross investment flow, tangible plus intangible acquisitions including contributions (INVAVAP), scaled by a perpetual-inventory capital stock. It is an alternative to the balance-sheet-change measure and gives the same pattern.

[23]The result holds under tighter identification. Across a ladder of fixed effects up to nonparametric country-specific exchange-rate slopes (Table C.10), the dominant cash-flow coefficient stays between 0.50 and 0.88 and

6.5 Heterogeneous real effects

Table 4 re-estimates the dominant-currency response within listing status, asset-size tertile, and the liquid and constrained tertiles of the Kaplan–Zingales index.[24] The theory predicts that the shock affects firms' real expenditure when external finance is costly at the margin ((29)). These measures are proxies for that shadow value. The real responses are aligned along such financing capacity. Cash flow is positive and significant in every subgroup (0.44–0.79), that is the valuation shock reaches all firm types. The compensation response is largest among small (0.27), constrained (0.31), and private (0.24) firms, and smallest among large (0.13) and public (0.12) firms, those with the most internal funds and possible financial hedges. Investment is flat in every subgroup except the most financially constrained tertile (0.11).

Table 4: Dominant-currency response by listing, size, and financial constraint

	Cash Flow (1)	Salaries (2)	Tot. CAPEX (3)
Listing			
Private	0.6920***	0.2422***	0.0394
	(0.1640)	(0.0804)	(0.0307)
Public	0.4373***	0.1208	-0.0538
	(0.1184)	(0.0697)	(0.1108)
Size			
Small	0.6788***	0.2707**	0.0140
	(0.1596)	(0.1058)	(0.0319)
Medium	0.6996***	0.2210***	0.0596
	(0.1773)	(0.0627)	(0.0652)
Large	0.5775***	0.1345*	0.0293
	(0.1477)	(0.0670)	(0.0991)
Financial constraint			
Liquid	0.7865***	0.1930**	-0.0489
	(0.1816)	(0.0698)	(0.0292)
Constrained	0.4923***	0.3066***	0.1072*
	(0.1351)	(0.0964)	(0.0579)

Notes: Coefficients on dominant-currency exposure for cash flow, salaries, and total CAPEX, each scaled by lagged total assets, estimated within subgroups defined one dimension at a time. Subgroups are listing status (private and public), asset-size tertile, and the liquid and constrained tertiles of the Kaplan–Zingales index. Estimates are within-model interactions. All controls are as in column (3) of Table 2. Standard errors in parentheses, double-clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

is significant at 1% in every column.

[24] The table reports the two extreme tertiles; the middle tertile is omitted.

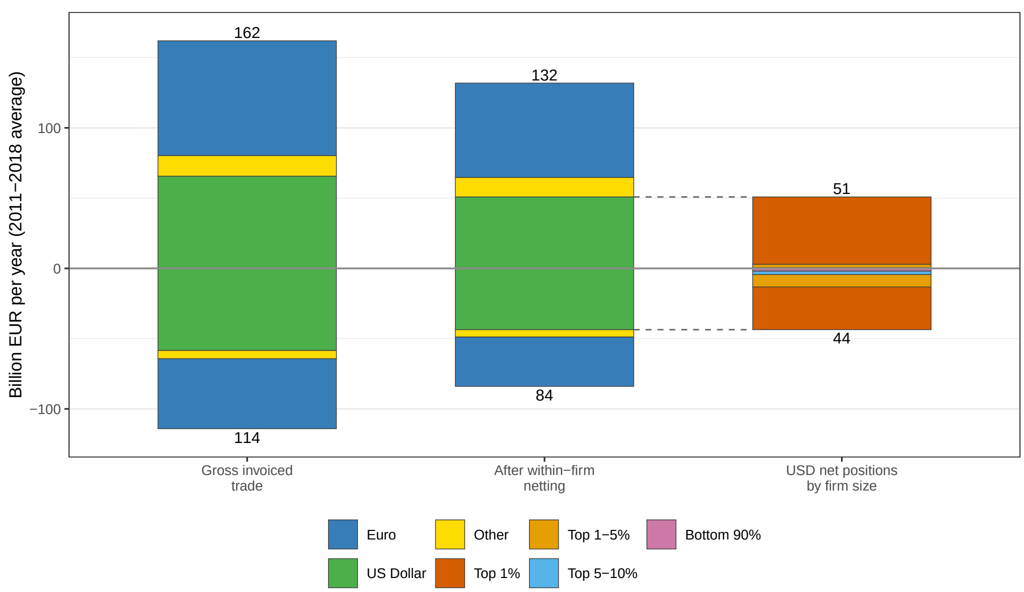
7 The Incidence of Valuation Shocks

The firm-level regressions identify a propensity to spend, or spending pass-through, out of invoice valuation shocks. They represent the response to one euro of invoice-currency mismatch, holding the amount of mismatch fixed. They say nothing about how much mismatch a firm carries. This section asks when the firm-level valuation effects aggregate into a large macroeconomic effect. Two components govern the answer, the pass-through estimates of section 6 and the distribution of the mismatch they apply to. Section 7.1 establishes the aggregate distribution. Section 7.2 combines exposure distribution and pass-through estimates to provide insights on the aggregate effects.

7.1 Who holds the exposure

Figure 4 traces the economy's aggregate trade in three stages, from gross flows, through within-firm netting, to the firm composition of the unhedged flows.

Figure 4: Within- and across-firm netting of invoice-currency trade, 2011–2018 averages



Notes: This figure reflects total French trade positions, as opposed to the positions in our sample in section 5 and section 6. The first bar shows gross invoiced exports (positive) and imports (negative) by invoice currency. The second bar shows the residual long and short positions after netting exports against imports *within each firm and currency*. Residual nets are grouped into euro, dollar, and other. The third bar splits the dollar net positions by exclusive bands of the within-year gross-trade distribution (top 1%, 1–5%, 5–10%, bottom 90%). Cumulative shares obtain by summing segments.

Gross extra-EU trade averages €162bn of exports and €114bn of imports per year in France (7.4% and 5.2% of GDP). The dollar-invoiced components are €66bn and €58bn (40.5% and

51.1% of exports and imports). Netting a firm's same-currency exports against its imports absorbs 23.8 percent of gross dollar trade (31.5% and 38.6% of total exports and imports are still unhedged within-firm). Natural hedging is important but partial and selective. As Figure 2 showed, it is how small and medium net exporters neutralize their exposure, while net importers stay short foreign currencies. These positions cancel only *across* firms, by 92.3 percent, leaving France's net dollar position at €7bn. The aggregate exposure is small even though the firms that make it up carry large individual positions.

The surviving exposure is highly concentrated (third bar in Figure 4). The top 1 percent of firms by gross trade hold 94.2 percent of the long and 69.8 percent of the short positions, and 148,244 firm-years are net short the dollar against 15,902 net long.[25] Note how the most intensely exposed firms, the small net importers (Figure 2), hold only a small share of the aggregate.[26] Section 7.2 will show that the fact that top traders carry most of France's unhedged position and have the lowest propensity to spend out of valuation income (Table 4) keeps the aggregate effect small for France.

7.2 The incidence accounting

This section provides a back-of-the-envelope accounting exercise to establish the magnitude of the aggregate response and to show that it is close to zero for several reasons.[27] Let X_g be the net USD-invoiced exposure (in euros) of firm group g.[28] Firms are partitioned by net-trade position and gross-trade size group. The propensities β_g^y are estimated within asset-size groups (Table 4). The accounting applies them to the customs universe binned by gross-trade size, mapping the two because the size rankings largely coincide.

[25]Large gross traders are also large in assets and employment, and signed dollar exposure rises with firm size on both margins (Table C.9 and Figure B.4), so the asset-size tertiles of Table 4 and the gross-trade bands of the accounting order firms similarly. The within-band exposure intensities of Table C.13, measured on the regression sample, reproduce the asset-tertile pattern of section 6.5.

[26]These figures cover extra-EU trade only. Intra-EU trade is overwhelmingly euro-invoiced.

[27]Two things are set aside on purpose. The first is the sampling uncertainty in the firm-level propensities β_g^y. The second is general-equilibrium feedback through prices, wages, and the exchange rate. Both are omitted because the conclusion concerns the sign and order of magnitude of the aggregate, which the accounting below shows to be small for reasons unrelated to the estimates' precision or to holding the rest of the economy fixed.

[28]Throughout the paper, *exposure* is measured on dollar invoicing, that is, every dollar-invoiced flow, including the share of dollar trade with the United States. This is the position that revalues when the euro/dollar rate moves. *Identification* of the invoicing effect, by contrast (the trade-relationship and firm-level pass-through that deliver β_g^y), uses dominant-currency pricing. This is dollar invoicing where the dollar is a vehicle currency, foreign to both France and the partner, which excludes dollar trade with the US. I estimate the pure valuation propensity off the clean DCP variation and apply it to the full USD position. Other foreign currencies are neglected. They invoice only 9.1% of exports and 5.2% of imports, against 40.5% and 51.1% for the dollar, so for France their omission is marginal.

The aggregate response to an exchange-rate movement $\Delta e^{€/\$}$ is

$$\Delta Y = \Delta e^{€/\$} \sum_g \beta_g^y X_g, \tag{41}$$

the group-by-group product of the net-exposure X_g and marginal propensity to spend out of valuation income for outcome y, β_g^y.

The aggregate effects depend on three features of this distribution, and the counterfactuals below switch them off one at a time. First, within-firm operational hedging still leaves about three-quarters of gross dollar trade exposed, and that surviving exposure is held by a handful of large traders, the top one percent (section 7.1). Second, the spending pass-through of valuation income is small in general. The mapped salary propensity to spend falls from 0.27 in the small-firm group to 0.13 in the large (Table 4), and it is close to zero for investment and employment. Third, propensity and exposure are inversely ordered, exposure is largest exactly where the propensity is smallest, so the high-propensity small importers hold only 4.5% of the short aggregate position. Table C.13 in the appendix provides more detail on exposure and propensity by net-trade position and size group.

Three counterfactuals isolate these forces. Table 5 evaluates all three for a 10% dollar appreciation, split into net exporters, net importers, and their sum. The **representative-trader** counterfactual switches off any excess of net-dollar exposure relative to gross trade. Within each net-trade position it spreads the net exposure across size groups in proportion to their gross trade, so every group carries the dollar-invoicing intensity of the representative trader conditional on its position being long or short, and with coefficients β_g^y held fixed. Because gross trade is itself concentrated among large traders, this counterfactual leaves that concentration largely in place, which is why its column moves little. The **all-constrained** counterfactual switches off low pass-through coefficients for large firms. It gives every group the propensity of the most financially constrained firms (the constrained Kaplan–Zingales tertile), with exposures held fixed. The **aligned** counterfactual switches off the misalignment between exposure and propensity. It keeps the actual exposures and the estimated propensities but swaps the small- and large-firm coefficients, so the groups that hold the most exposure also carry the highest propensity.

In the data, long and short dollar positions are of roughly equivalent magnitude and move in opposite directions. Therefore, they nearly cancel when summed in the last panel across all variables and across all counterfactuals. The net-exporter cash-flow response of €2.94bn (about 0.13 percent of GDP) and the net-importer response of €-2.57bn (-0.12 percent) leave a net €0.37bn, some 0.02 percent of GDP, and the net salary response is €0.05bn, roughly 0.002 percent. Note, however, that this is an across-firms cancellation. The second cancella-

Table 5: Actual and counterfactual responses to a 10% dollar appreciation

Outcome	Actual $\sum_g \beta_g X_g \Delta e$	Counterfactuals: Representative trader $X_g/\mathrm{trade}_g = \bar{x}$	All constrained $\beta_g = \beta_{\mathrm{constr}}$	Aligned $\beta_{\mathrm{small}} \rightleftarrows \beta_{\mathrm{large}}$
Net exporters (net long USD)				
Cash flow	2.94	2.98	2.50	3.45
Salaries	0.69	0.73	1.56	1.37
Total CAPEX	0.15	0.15	0.54	0.07
Net importers (net short USD)				
Cash flow	-2.57	-2.55	-2.15	-2.95
Salaries	-0.63	-0.62	-1.34	-1.14
Total CAPEX	-0.13	-0.13	-0.47	-0.07
Net (exporters + importers)				
Cash flow	0.37	0.42	0.36	0.51
Salaries	0.05	0.10	0.22	0.23
Total CAPEX	0.02	0.02	0.08	-0.00

Notes: Responses in billion euro per year of each outcome to a 10% dollar appreciation, computed from (41) and split into net exporters, net importers, and their sum (the Net panel). MPX denotes the marginal propensity to spend out of valuation income. The propensities are the size-tertile estimates of Table 4, identified from dominant-currency variation on the regression sample (2001–2019) and applied to both net-trade positions within a size group. Each propensity multiplies the net USD-invoiced exposure of its net-position and gross-trade size-group cell in the customs universe (2011–2018 averages), as explained in section 7.2. **Representative trader** spreads each position's net exposure across its groups in proportion to their gross trade, holding propensities fixed. **All constrained** applies the propensity of the constrained Kaplan–Zingales tertile to every group. **Aligned** keeps the exposures and the estimated propensities but swaps the small- and large-firm coefficients, so the highest-exposure group carries the highest propensity. For scale, French nominal GDP averaged about €2,190 bn over 2011–2018, so the net responses are about 0.02 percent (cash flow) and 0.002 percent (salaries) of GDP, and each net-trade position separately is on the order of 0.1 percent.

tion operates within each group position (net exporters and net importers in isolation). Even within firms with the same position, the surviving exposure sits with the low-propensity firms. Reversing this misalignment is what the aligned counterfactual measures, and it lifts the net response to €0.23bn, 4.2 times the actual. The within-position order is therefore a large force in proportional terms and itself sets which firms bear the incidence.

The near-zero aggregate is an incidence result, not a finding of neutrality. Among net importers, dollar exposure relative to sales is largest for the smallest and most financially constrained firms, and these firms bear the largest payroll response, and at the margin investment response, per euro of sales. They weigh little in the aggregate only because each is small in nominal terms. Yet they are numerous, together employing roughly a third of France's workforce among trading firms in the sample. Moreover, absorbing a valuation shock as liquidity is not free for large firms because the liquidity and hedges that keep the shock off payroll are themselves costly to hold.

One final caveat on who bears the incidence in this context. The accounting places each euro of incidence on the firm that holds the dollar invoice. However, wholesale-type importers, who buy in dollars to resell at home, may pass part of the cost on to domestic buyers. Table C.14 classifies 56.5% of net importers as wholesale-type and the remaining 43.5% as producers, and the exposed population spans manufacturing, wholesale, construction, and other sectors (Table C.15). Such pass-through to consumers would reassign the valuation loss within the French economy, from the trader to other domestic actors, and I do not estimate such customers' spending response. A downstream propensity in the range of the estimated ones would leave the net aggregate of Table 5 unchanged in sign and order of magnitude.

8 Conclusion

In the short run an exchange-rate movement changes the value of trade far more than its volume, so a firm's exposure is set by the currency its operations are priced in and appears as a cash-flow shock. Measuring exposure by invoice currency rather than trade weights recovers a cash-flow shock three to five times larger than any effective-rate-based measure. That shock passes into cash flow at a high but incomplete rate, around half to two-thirds, but firms retain about 70 percent of it as cash and working capital, and it reaches payroll and investment only for firms with scarce internal funds. The aggregate cash-flow effect is small, on the order of 0.02 percent of GDP. Dollar exposure cancels between long and short firms, and the exposure that survives within-firm netting is concentrated among large, liquid firms least likely to spend it. In the short term, exchange-rate disconnect is thus an incidence outcome, governed by who holds the valuation exposure. The same design recovers larger real effects in more exposed economies (Nookhwun et al. 2025). Where this kind of shocks are severe enough, firms switch invoice currency itself (Garofalo et al. 2024, Cristoforoni and Errico 2024).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Claude Code (Anthropic) and Refine to improve the content, language, and readability of the manuscript. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article.

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A Identification: The Shift-Share Estimator as a Year-Level Instrument

The dominant-currency exposure in (39) is a shift-share regressor, a predetermined net dollar position (the share) revalued by the euro-dollar movement (the shock). Borusyak et al. (2022) show that such an estimator can be read as a just-identified instrumental-variable regression run at the level of the shocks, here the year. This appendix states the equivalence in the paper's notation and draws out its implications for the variation behind $\beta^{\$}$ and for inference.

Write the dollar channel as a predetermined (lagged) share times the common shock,

$$\widetilde{I}^{\$}_{ft} = s^{\$}_{f,t}\,\Delta e^{€/\$}_{t}, \qquad s^{\$}_{f,t} = \frac{N^{I,\$}_{f,t-1}}{A_{f,t-1}}, \tag{A.1}$$

the net dollar position of (35) scaled by lagged assets, the same denominator as the main firm regression (39), so the year-level ratio (A.2) is the exact Frisch–Waugh representation of that regression rather than an approximation.[29] Let $\widetilde{Y}^{\perp}_{ft}$ and $\widetilde{I}^{\$\perp}_{ft}$ denote the outcome and the dollar index after partialling out the controls and fixed effects of (39). The estimator then collapses to a year-level ratio,

$$\hat{\beta}^{\$} = \frac{\sum_{ft}\widetilde{I}^{\$\perp}_{ft}\,\widetilde{Y}^{\perp}_{ft}}{\sum_{ft}\widetilde{I}^{\$\perp}_{ft}\,\widetilde{I}^{\$\perp}_{ft}} = \frac{\sum_{t}\Delta e^{€/\$}_{t}\,\widehat{Y}^{\perp}_{t}}{\sum_{t}\Delta e^{€/\$}_{t}\,\widehat{I}^{\$\perp}_{t}}, \qquad \widehat{Y}^{\perp}_{t} = \sum_{f} s^{\$}_{f,t}\,\widetilde{Y}^{\perp}_{ft}, \tag{A.2}$$

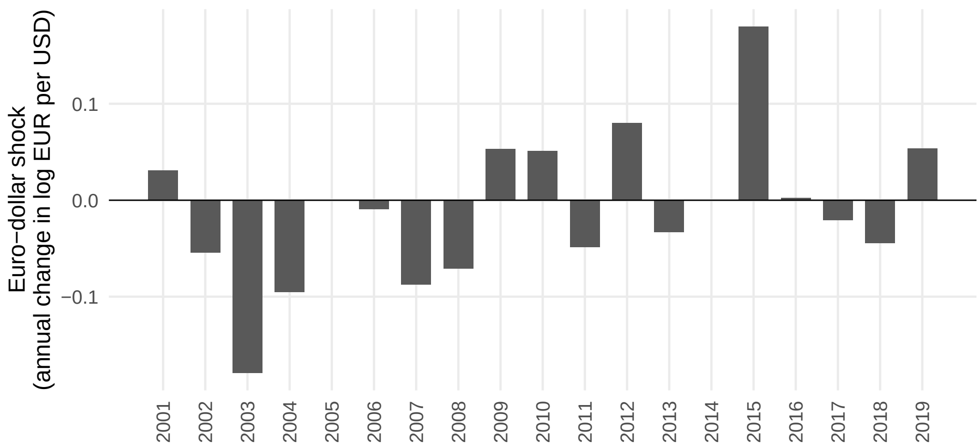
with $\widehat{I}^{\$\perp}_{t}$ defined analogously. The last expression is an instrumental-variable estimator in which the exposure-weighted aggregate outcome $\widehat{Y}^{\perp}_{t}$ is projected on the exposure-weighted aggregate index, with the euro-dollar movement $\Delta e^{€/\$}_{t}$ as the instrument. The identifying variation is at the yearly level, fewer than twenty annual euro-dollar shocks over the sample.

Effective shock variation. Because the estimator runs on this short series of annual shocks, a natural concern is that one or two large swings carry the result. Two features of the series address it. First, the euro-dollar episodes do not line up with France's own macroeconomic or financial events. The swings track a global dollar cycle driven by forces external to France

[29]Because the instrument $\Delta e^{€/\$}_{t}$ is common to all firms within a year, the year-level structure below is unaffected by the share being updated each year. The loading $s^{\$}_{f,t}$ is predetermined relative to the within-year shock.

(Obstfeld and Zhou 2022), so the variation behind $\beta^{\$}$ is not a relabeled domestic cycle. Second, the shocks satisfy the requirement of the year-level instrument that they not be autocorrelated. Over the 2001–2019 window the annual euro-dollar movements alternate in sign (Figure A.1), and their first-order autocorrelation is 0.17, statistically indistinguishable from zero. No single year dominates (the largest accounts for under a third of the squared variation), and the two largest swings, the dollar depreciation of 2003 and the surge of 2014–15, are of almost equal size and opposite sign, so identification is not a single directional episode.

Figure A.1: Annual euro-dollar shocks over the estimation window

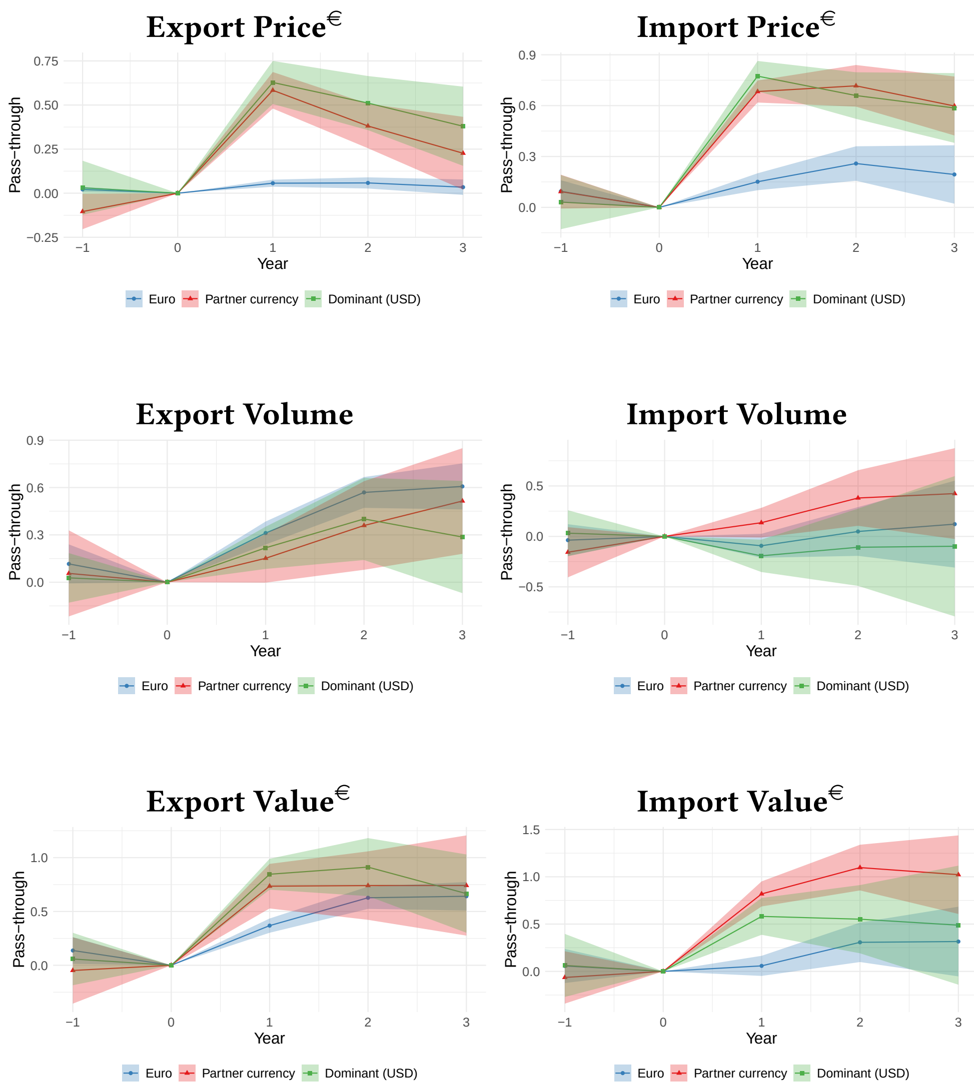


Notes: The yearly change in the log euro value of a dollar, $\Delta e_t^{€/\$}$, over 2001–2019, from annual-average rates. This is the shift component of the dominant-currency index and the shock series the year-level instrument of (A.2) aggregates. The movements alternate in sign, with a first-order autocorrelation of 0.17 ($p = 0.50$).

Inference under a common shock. The year-level representation also pins down the level at which to cluster. Because the identifying variation is the euro-dollar shock common to all firms in a year, the exposure-robust inference of Adão et al. (2019) for shift-share designs corresponds, in this just-identified case, to clustering at the year (shock) level. The estimates in the paper are double-clustered by firm and year. They retain the year dimension that carries the common shock and allow within-firm serial correlation on top. Multiway clustering is not in general more conservative than clustering on a single dimension, so this is a reporting choice rather than a bound. In this setting, however, I computed the Adão et al. (2019) exposure-robust standard errors separately and verified that the firm-and-year double-clustered standard errors are the more conservative of the two. This is why the firm specifications cluster on both margins.

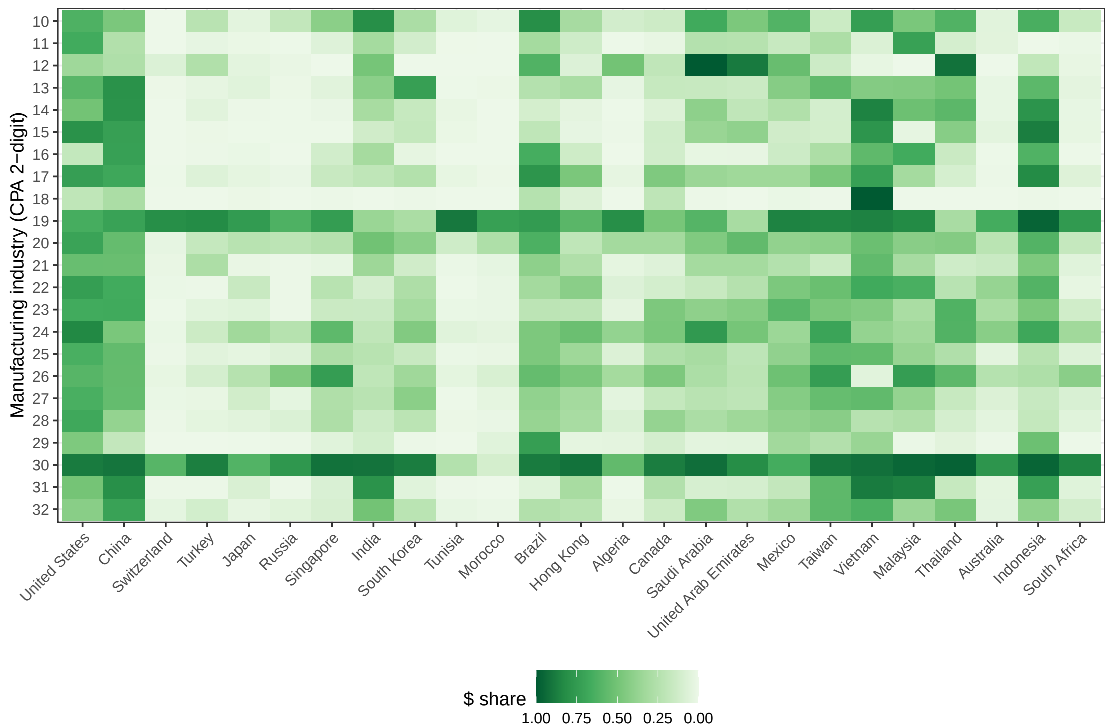
B Additional Figures

Figure B.1: Exchange-rate pass-through by invoicing regime, annual impulse responses



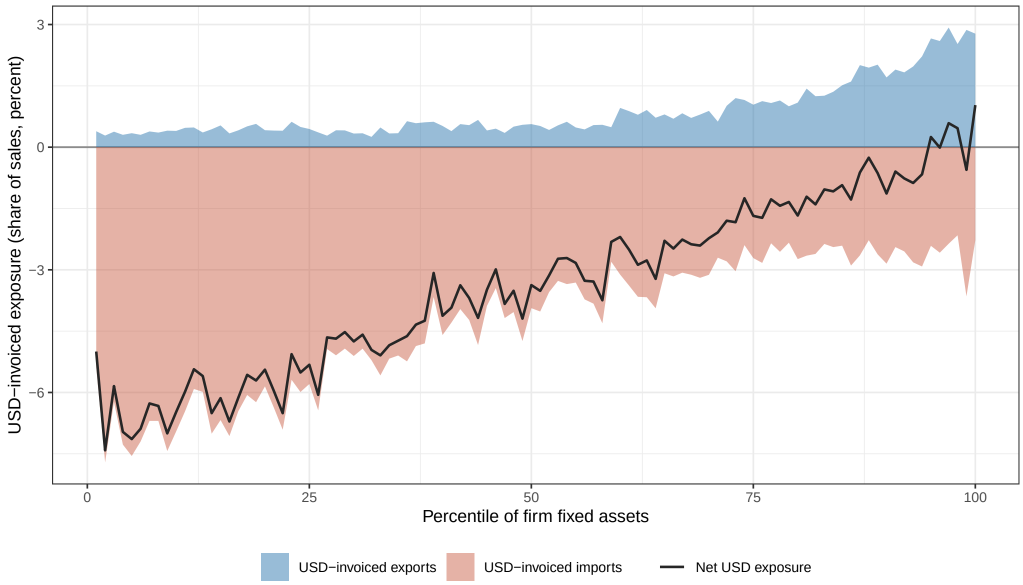
Notes: Annual counterpart of Figure 1. Local-projection impulse responses, (33) at annual frequency over 2000–2018, of the log euro unit value (price), log real volume, and log euro value of a trade relationship to a euro depreciation, by invoicing regime (euro, partner currency, and dominant dollar). Each panel is one outcome and one trade direction, exports left and imports right. The plotted dominant coefficient is the euro/dollar leg. All panels include relationship and year fixed effects, with trade-weighted partner-country CPI and nominal GDP as controls. Standard errors are double-clustered by relationship and year. This relationship-level exercise is distinct from the product-level annual pass-through tables (Tables C.5 and C.6), which use product and time fixed effects. Shaded areas are 95% confidence intervals.

Figure B.2: Dollar-invoicing share by partner country and manufacturing industry, 2011–2018



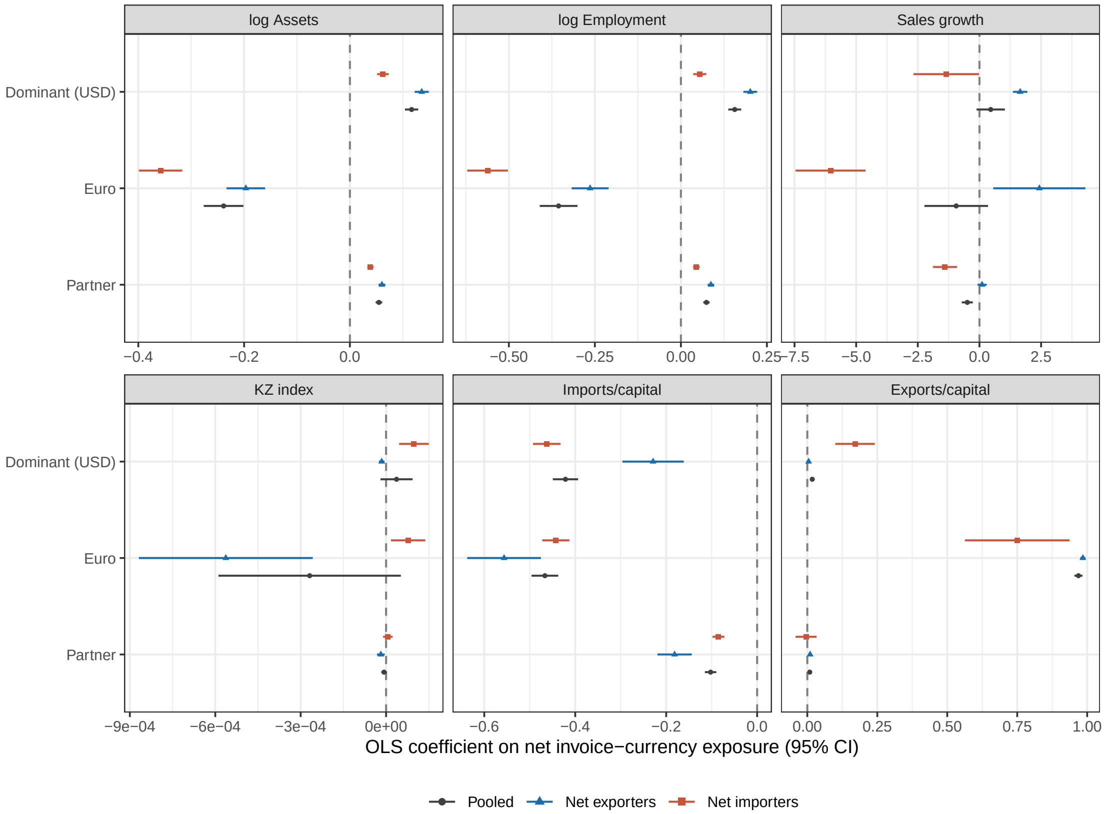
Notes: Rows are 2-digit manufacturing industries (CPA classification) and columns are partner countries. Each cell is the dollar-invoiced share of the industry-country pair's extra-EU trade value over 2011–2018. Darker cells are more dollarized.

Figure B.3: Net USD-invoiced exposure by firm size, pooled across net-trade positions



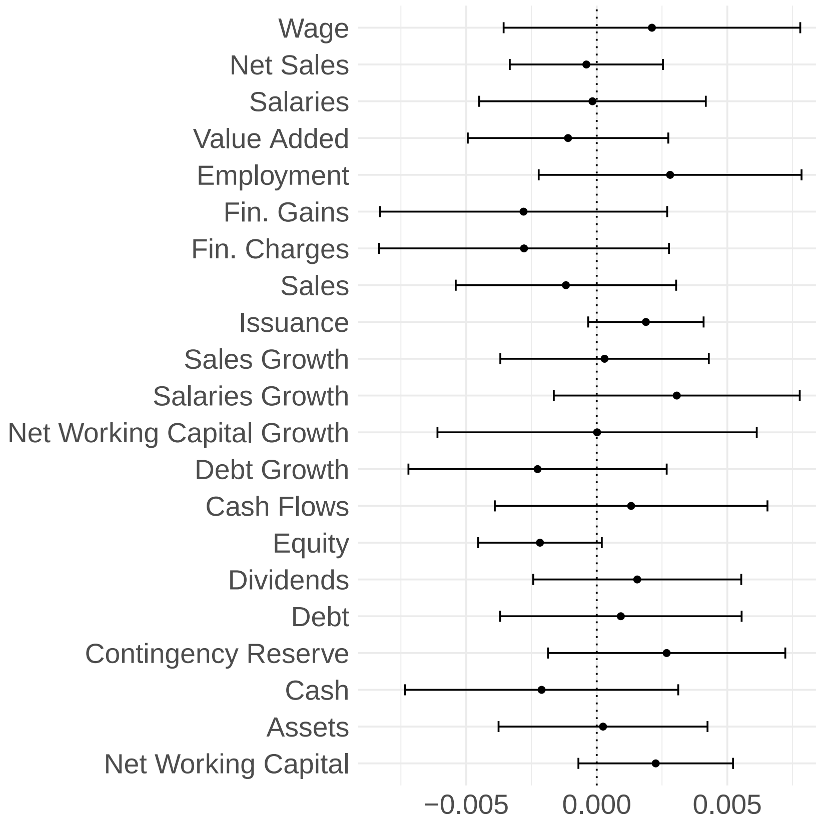
Notes: Pooled counterpart of Figure 2, without the net-trade-position split. Firm-years are ranked into percentiles of fixed assets. Shaded areas are mean USD-invoiced exports (positive) and imports (negative) over sales. The black line is the mean net position.

Figure B.4: Firm characteristics and net invoice-currency exposure



Notes: Each point is an OLS coefficient, with its 95% confidence interval, of net dominant-currency share over capital on the firm characteristic.

Figure B.5: Balance test of the dominant-currency index against lagged firm outcomes



Notes: Each point is the correlation, with its 95% confidence interval, of the standardized dominant-currency valuation index with a residualized, standardized lagged balance-sheet or income-statement variable (cash flow, salaries, value added, sales, investment, net working capital, debt, issuance, and their growth rates). The correlations test the shift-share identifying assumption that the valuation shock is orthogonal to firms' pre-shock trajectories. Each lagged outcome is residualized on the controls and fixed effects of the benchmark firm specification in Table 3 before being correlated with the index. The dotted line marks zero. Standard errors are double-clustered by firm and year.

C Additional Tables

Table C.1: Sample coverage of extra-EU French trade

	Full trade sample	Firm-level (CASD) sample
Firms	351,042	19,115
Share of extra-EU exports	100%	61.5%
Share of extra-EU imports	100%	60.0%
Years	2011–2018	2001–2019
Balance sheets	—	FARE/FICUS

Notes: The left column represents the full universe of extra-EU French customs trade. The right column represents the sample used throughout the paper, firms that trade outside the EU in every year of the customs window, 2000–2018, and whose customs records merge with FARE/FICUS balance sheets. The firm-level regression sample drawn from the right column is followed over the balance-sheet panel, 2001–2019. The restriction leaves the bulk of trade value in the sample. The full trade sample is used for the aggregate accounting of section 7, after partial-equilibrium identification on the paper's sample.

Table C.2: Yearly transition matrix of invoicing regimes (product level)

	Euro	Partner	Dominant
Euro	98.2%	0.7%	1.1%
Partner	3.7%	95.6%	0.7%
Dominant	3.3%	0.5%	96.2%

Notes: Year-to-year transition matrix of the invoicing regime (euro, partner, or dominant currency) of a product, defined as a firm $\times$ HS6 $\times$ partner $\times$ shipment-terms combination, over 2011–2018. The sample is restricted to single-currency products, those never invoiced in more than one currency within a year. Each entry is the share of product-year transitions from the row regime at year $t-1$ to the column regime at year t.

Table C.3: Annual pass-through by invoicing regime with observed invoicing, 2011–2018

	Imports			**Exports**		
	Price (1)	Quantity (2)	Value (3)	Price (4)	Quantity (5)	Value (6)
Euro $\times \Delta e(€/*)$	0.0940	-0.1455	-0.0515	0.0291	0.3895***	0.4187***
	(0.0632)	(0.1371)	(0.1270)	(0.0207)	(0.0528)	(0.0545)
Partner $\times \Delta e(€/*)$	0.7777***	0.1552	0.9330***	0.7245***	0.1140	0.8385***
	(0.0439)	(0.1306)	(0.1451)	(0.0645)	(0.1279)	(0.1611)
Dominant $\times \Delta e(€/\$)$	0.8663***	-0.1446	0.7216***	0.7610***	-0.0953	0.6657**
	(0.0852)	(0.1346)	(0.1527)	(0.0968)	(0.1333)	(0.1976)
Dominant $\times \Delta e(*/\$)$	-0.1736*	0.2266	0.0530	-0.1020*	-0.1859	-0.2879*
	(0.0715)	(0.1477)	(0.1710)	(0.0451)	(0.1352)	(0.1282)
Observations	583,849	583,849	583,849	1,160,267	1,160,267	1,160,267
R^2	0.13370	0.08586	0.15663	0.10583	0.09169	0.09778
Within R^2	0.07943	0.05718	0.08406	0.10416	0.06846	0.06453
Product fixed effects	✓	✓	✓	✓	✓	✓
Time fixed effects	✓	✓	✓	✓	✓	✓

Notes: As Table C.6, restricted to 2011–2018, the years with observed invoicing currency and no imputation, and likewise estimated on a balanced product-level panel. A product is defined as a firm $\times$ HS6 product code $\times$ partner country $\times$ invoice regime $\times$ flow combination. Standard errors in parentheses, double-clustered by product and time. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Euro value pass-through under alternative invoice-share constructions

	Exports			Imports		
	Pooled (1)	2011 base (2)	Leave-one-out (3)	Pooled (4)	2011 base (5)	Leave-one-out (6)
Euro $\times \Delta e(€/*)$	0.3373***	0.3535***	0.3753***	-0.0765	-0.0459	-0.0622
	(0.0398)	(0.0433)	(0.0422)	(0.0442)	(0.0492)	(0.0573)
Partner $\times \Delta e(€/*)$	0.7346***	0.7044***	0.4612***	0.7893***	0.7733***	0.7833***
	(0.1002)	(0.1247)	(0.0956)	(0.0592)	(0.0690)	(0.0903)
Dominant $\times \Delta e(€/\$)$	0.8116***	0.7882***	0.6113***	0.4908***	0.4573***	0.4391***
	(0.0688)	(0.0781)	(0.0819)	(0.0791)	(0.0822)	(0.0877)
Dominant $\times \Delta e(*/\$)$	-0.3861***	-0.3775***	-0.2914**	0.0620	0.1110	0.0028
	(0.0872)	(0.0896)	(0.1100)	(0.0874)	(0.0863)	(0.1090)
Observations	1,739,839	1,433,070	1,792,744	911,044	768,465	959,722

Notes: Annual euro value pass-through (the value column of Table C.6) on the balanced product-level long panel, 2000–2018, under three invoice-currency-share constructions. *Pooled* is the baseline value share over 2011–2018 of (32). *2011 base* uses only the first observed year, the construction that minimizes look-ahead. *Leave-one-out* replaces a relationship's own share by the product$\times$country$\times$currency cell average. All columns include product and time$\times$transaction-length fixed effects. Standard errors in parentheses, double-clustered by product and time. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.5: Fixed-effect ladder for the annual value pass-through by invoicing regime

	(1)	(2)	(3)	(4)
Panel A: Exports Value				
Euro $\times \Delta e(€/*)$	0.4632***	0.3831***	0.3373***	0.0028
	(0.1059)	(0.0363)	(0.0398)	(0.0700)
Partner $\times \Delta e(€/*)$	0.7356***	0.7607***	0.7346***	0.3627***
	(0.1867)	(0.1053)	(0.1002)	(0.1000)
Dominant $\times \Delta e(€/\$)$	0.7522***	0.8491***	0.8116***	0.4511***
	(0.1614)	(0.0753)	(0.0688)	(0.0460)
Panel B: Imports Value				
Euro $\times \Delta e(€/*)$	0.1782	0.0427	-0.0765	-0.0599
	(0.1266)	(0.0560)	(0.0442)	(0.0925)
Partner $\times \Delta e(€/*)$	0.6476***	0.8548***	0.7893***	0.6419***
	(0.1439)	(0.0670)	(0.0592)	(0.1456)
Dominant $\times \Delta e(€/\$)$	0.4656**	0.5963***	0.4908***	0.5592***
	(0.1678)	(0.1003)	(0.0791)	(0.1203)
Product FE	✓	✓	✓	✓
Time FE		✓		
Time×Length FE			✓	✓
Nonparam. country ER slopes				✓

Notes: Euro value response to the exchange-rate leg of each invoicing regime (euro, partner, and dominant), long panel with imputed invoicing before 2011, balanced sample, 2000–2018. Panel A is exports and Panel B imports. Columns add fixed effects from left to right, from product (1), to product plus time (2), to product plus time×transaction-length (3), the Table C.6 benchmark, to a nonparametric exchange-rate slope per partner country (4). The partner-dollar leg is included as a control but is absorbed by collinearity in column (4). Standard errors in parentheses, double-clustered by product and time. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6: Annual pass-through by invoicing regime, 2000–2018

	Imports			**Exports**		
	Price (1)	Quantity (2)	Value (3)	Price (4)	Quantity (5)	Value (6)
Euro $\times \Delta e(€/*)$	0.1404***	-0.2169***	-0.0765	0.0200*	0.3173***	0.3373***
	(0.0338)	(0.0553)	(0.0442)	(0.0101)	(0.0396)	(0.0398)
Partner $\times \Delta e(€/*)$	0.6822***	0.1071	0.7893***	0.5660***	0.1686**	0.7346***
	(0.0386)	(0.0744)	(0.0592)	(0.0484)	(0.0771)	(0.1002)
Dominant $\times \Delta e(€/\$)$	0.7620***	-0.2712***	0.4908***	0.6035***	0.2082**	0.8116***
	(0.0546)	(0.0762)	(0.0791)	(0.0583)	(0.0736)	(0.0688)
Dominant $\times \Delta e(*/\$)$	-0.2382***	0.3001***	0.0620	-0.0983**	-0.2878***	-0.3861***
	(0.0355)	(0.0745)	(0.0874)	(0.0394)	(0.0762)	(0.0872)
Observations	911,044	911,044	911,044	1,739,839	1,739,839	1,739,839
R^2	0.13844	0.10187	0.08366	0.07021	0.09832	0.10513
Within R^2	0.09616	0.09185	0.07584	0.01370	0.04445	0.09245
Product fixed effects	✓	✓	✓	✓	✓	✓
Time fixed effects	✓	✓	✓	✓	✓	✓

Notes: Annual euro price, quantity, and value responses to the exchange-rate leg of each invoicing regime (euro, partner, and dominant), baseline long panel with imputed invoicing before 2011, estimated on a balanced panel. Standard errors in parentheses, double-clustered by product and time. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.7: Relationship entry and exit by invoicing regime, 2011–2018

	Imports		**Exports**	
	Entry	Exit	Entry	Exit
	(1)	(2)	(3)	(4)
Euro $\times \Delta e(€/*)$	0.0425	0.0308	0.0107	-0.0661
	(0.0439)	(0.0880)	(0.0199)	(0.0361)
Partner $\times \Delta e(€/*)$	0.0740	-0.1293	-0.0701	0.3499
	(0.0751)	(0.2726)	(0.1125)	(0.7147)
Dominant $\times \Delta e(€/\$)$	0.0614	-0.2109	0.0252	-0.3742
	(0.0729)	(0.1938)	(0.0432)	(0.2412)
Dominant $\times \Delta e(*/\$)$	0.0078	0.0386	0.0507*	-0.2796**
	(0.0585)	(0.1585)	(0.0230)	(0.0967)
Observations	5,436,689	2,405,098	9,733,245	3,798,732

Notes: The benchmark specification of Table C.6 re-estimated with relationship entry and exit as outcomes in place of price, quantity, and value, over the window in which invoicing is observed. *Entry* is a linear-probability indicator, among relationship-years not traded in year $t-1$ for being traded in year t. *Exit* is a linear-probability indicator, among relationship-years traded in year $t-1$, for not being traded in year t. The regressors are the same invoicing-share$\times$exchange-rate interactions as in Table C.6, with partner-country CPI and nominal GDP as controls. All columns include product and year fixed effects. Standard errors in parentheses, double-clustered by relationship and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.8: Annual pass-through with currency-choice determinants as controls

	Imports			**Exports**		
	Price (1)	Quantity (2)	Value (3)	Price (4)	Quantity (5)	Value (6)
Euro $\times \Delta e(€/*)$	0.1433***	-0.2098***	-0.0665	0.0199*	0.3188***	0.3388***
	(0.0335)	(0.0547)	(0.0430)	(0.0102)	(0.0401)	(0.0403)
Partner $\times \Delta e(€/*)$	0.6786***	0.0993	0.7779***	0.5645***	0.1487**	0.7131***
	(0.0384)	(0.0737)	(0.0582)	(0.0499)	(0.0691)	(0.0945)
Dominant $\times \Delta e(€/\$)$	0.7672***	-0.2699***	0.4973***	0.5955***	0.2072**	0.8028***
	(0.0536)	(0.0768)	(0.0802)	(0.0621)	(0.0749)	(0.0648)
Dominant $\times \Delta e(*/\$)$	-0.2510***	0.2947***	0.0437	-0.0922**	-0.2794***	-0.3716***
	(0.0354)	(0.0760)	(0.0873)	(0.0387)	(0.0753)	(0.0826)
Size $\times \Delta e(€/*)$	-0.0161***	0.0179	0.0018	-0.0013	0.0230*	0.0217*
	(0.0044)	(0.0140)	(0.0145)	(0.0023)	(0.0109)	(0.0111)
Mkt. share $\times \Delta e(€/*)$	0.0670***	0.0619	0.1289**	0.0305	-0.0803	-0.0498
	(0.0228)	(0.0589)	(0.0585)	(0.0288)	(0.0640)	(0.0743)
Intensity $\times \Delta e(€/*)$	-0.1747*	-0.2926	-0.4673*	-0.0809	0.6730**	0.5921**
	(0.0881)	(0.2597)	(0.2220)	(0.0573)	(0.2305)	(0.2656)
Observations	911,044	911,044	911,044	1,739,839	1,739,839	1,739,839
R^2	0.14492	0.13383	0.08330	0.08440	0.13662	0.18418
Within R^2	0.10018	0.11393	0.06032	0.05703	0.10242	0.16808
Product fixed effects	✓	✓	✓	✓	✓	✓
Time fixed effects	✓	✓	✓	✓	✓	✓

Notes: As Table C.6 (same balanced product-level panel, fixed effects, and clustering), adding firm size, market share, and imported-input intensity defined as in Amiti et al. (2022), each interacted with the euro/partner exchange-rate change, as controls. The controls are centered at their sample mean. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.9: Firm characteristics, dollar exposure, and financing

	Net exporters			Net importers		
	Small	Medium	Large	Small	Medium	Large
Trade and net dollar exposure (percent of sales)						
Imports	0.00	0.08	0.36	20.9	13.5	7.14
Exports	13.7	8.24	8.34	0.56	0.95	0.68
Net USD exposure (mean)	-0.06	0.16	1.09	-12.6	-9.45	-6.07
Firm characteristics (percent of fixed capital unless noted)						
Employment (headcount)	5.00	24.0	125	6.00	26.0	131
Sales	9.93	3.63	1.89	12.4	5.83	2.36
Cash holdings	54.5	16.7	4.01	70.4	23.0	4.57
Debt to suppliers	124	46.6	24.9	148	71.8	31.0
Interest charged	1.38	1.15	0.79	2.34	1.82	0.93
Firms	3,987	4,691	4,990	3,562	3,454	3,277
Firm-years	43,681	52,816	59,638	39,041	32,887	28,476

Notes: Cells are within-group medians, except the net-USD-exposure row, which is a mean, since the net-exposure median is zero in most cells. Trade flows and the net-USD-exposure row are percent of sales. The firm-characteristics panel is percent of fixed capital, except employment (headcount). The net-USD-exposure row is the firm-count-weighted tertile mean of the sales-normalized net dollar-invoiced position of Figure 2, whose fixed-asset ranking runs within each net-position group. Its size tertiles are therefore within-group while the other rows use pooled asset-size tertiles. The Firms row counts the distinct firms in each cell's regression sample. Firms that switch net-trade position appear in both halves. Firm-years are from the descriptive sample.

Table C.10: Fixed-effect ladder for FX exposure and firm outcomes

	(1)	(2)	(3)	(4)
Panel A: Cash Flow				
Euro exposure, $I^{€}_{ft}$	-0.0073	0.0228	0.0282	0.0042
	(0.0244)	(0.0341)	(0.0316)	(0.0366)
Partner exposure, I^{*}_{ft}	0.5629***	0.3769***	0.3703***	0.4169***
	(0.1067)	(0.0728)	(0.0789)	(0.1287)
Dominant exposure, $I^{\$}_{ft}$	0.8778***	0.6864***	0.6638***	0.5003***
	(0.2406)	(0.1734)	(0.1565)	(0.0908)
Panel B: Salaries				
Euro exposure, $I^{€}_{ft}$	-0.0222	-0.0061	-0.0024	0.0028
	(0.0258)	(0.0148)	(0.0136)	(0.0178)
Partner exposure, I^{*}_{ft}	0.1837**	0.0688	0.0966	-0.0169
	(0.0796)	(0.0572)	(0.0581)	(0.0677)
Dominant exposure, $I^{\$}_{ft}$	0.3113**	0.2054***	0.2234***	0.1378*
	(0.1116)	(0.0634)	(0.0697)	(0.0648)
Panel C: Total CAPEX				
Euro exposure, $I^{€}_{ft}$	-0.0315**	-0.0249	-0.0250	-0.0195
	(0.0124)	(0.0153)	(0.0154)	(0.0179)
Partner exposure, I^{*}_{ft}	0.0381	-0.0113	-0.0295	-0.0211
	(0.0740)	(0.0811)	(0.0796)	(0.1292)
Dominant exposure, $I^{\$}_{ft}$	0.1422*	0.0481	0.0474	-0.0195
	(0.0761)	(0.0416)	(0.0422)	(0.0336)
Firm FE	✓	✓	✓	✓
Year FE		✓		
Industry-Year FE			✓	✓
Nonparam. country slopes				✓

Notes: Coefficients on euro, partner-currency, and dominant (dollar) exposure for cash flow, salaries, and total CAPEX, each scaled by lagged total assets, under an increasing set of fixed effects across columns. Column (3), with firm and industry×year fixed effects, is the main-text benchmark specification. Column (4) adds nonparametric country-specific exchange-rate slopes. Standard errors in parentheses, double-clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.11: Firm-level responses with exposure fixed at base period, across sample windows

	2001–2016 (1)	2001–2019 (2)	2001–2021 (3)
Panel A: Cash Flow			
Euro exposure, $I^{€}_{ft}$	0.0600*	0.0579**	0.0666**
	(0.0281)	(0.0251)	(0.0244)
Partner exposure, I^{*}_{ft}	0.1916**	0.2131***	0.2367***
	(0.0770)	(0.0638)	(0.0625)
Dominant exposure, $I^{\$}_{ft}$	0.6316***	0.5546***	0.5622***
	(0.1243)	(0.1401)	(0.1422)
Panel B: Salaries			
Euro exposure, $I^{€}_{ft}$	0.0104	0.0132	0.0159
	(0.0109)	(0.0106)	(0.0097)
Partner exposure, I^{*}_{ft}	0.0631	0.0621*	0.0492
	(0.0419)	(0.0348)	(0.0333)
Dominant exposure, $I^{\$}_{ft}$	0.1661***	0.1344**	0.1200**
	(0.0549)	(0.0515)	(0.0553)
Panel C: Total CAPEX			
Euro exposure, $I^{€}_{ft}$	-0.0145	-0.0115	-0.0085
	(0.0121)	(0.0096)	(0.0099)
Partner exposure, I^{*}_{ft}	-0.0559	-0.0426	-0.0457
	(0.0562)	(0.0509)	(0.0503)
Dominant exposure, $I^{\$}_{ft}$	0.0164	0.0072	0.0018
	(0.0301)	(0.0289)	(0.0282)
Observations	162,463	205,065	231,892

Notes: Coefficients on dominant, partner, and euro net invoice-currency exposure for cash flow, salaries, and total CAPEX. The benchmark specification (39) updates net positions each year and scales them by lagged total assets. This table tests a fixed-time alternative: net positions are held at the 2000 base period and scaled by base-year assets, so a firm's exposure varies only through the exchange rate. The outcomes remain scaled by lagged total assets. Columns move the panel's end-year from 2016 to 2019 to 2021. All controls are as in column (3) of Table 2. Standard errors in parentheses, double-clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.12: Dominant-currency response of cash flow and financial income by listing and size

	Cash Flow (1)	Financial Income (2)
Listing		
Private	0.6920***	-0.0021
	(0.1640)	(0.0154)
Public	0.4373***	-0.0887**
	(0.1184)	(0.0332)
Size		
Small	0.6788***	0.0175
	(0.1596)	(0.0213)
Medium	0.6996***	-0.0320
	(0.1773)	(0.0184)
Large	0.5775***	-0.0523*
	(0.1477)	(0.0252)

Notes: Coefficients on dominant (dollar) net invoice-currency exposure for cash flow and for financial income, each scaled by lagged total assets, by listing status and asset-size tertile. Estimates are as in Table 4. All controls are as in column (3) of Table 2. Standard errors in parentheses, double-clustered by firm and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.13: Exposure holdings, intensity, and propensity by net USD position and size group

Size (gross trade)	Firm-years per year	Gross trade (bn EUR/yr)	Net exposure (bn EUR/yr)	Net/gross (%)	Net/sales (%)	MPX salaries β^{sal} (s.e.)
Net exporters (net long USD)						
Small	2,316	0.5	0.1	27.1	0.1	0.27 (0.11)
Medium	871	1.4	0.3	22.2	1.9	0.22 (0.06)
Large	1,867	134.7	50.4	37.4	4.1	0.13 (0.07)
Net importers (net short USD)						
Small	43,012	4.2	-2.0	-46.3	-8.4	0.27 (0.11)
Medium	3,988	6.3	-2.4	-38.3	-15.5	0.22 (0.06)
Large	4,176	108.1	-39.2	-36.3	-13.9	0.13 (0.07)

Notes: Firm-years are grouped by the sign of their net USD-invoiced exposure (dollar invoice currency, including dollar trade with the US) and by within-year gross-trade size group. Large consists of firms at the top 5% of gross traders, medium the 5–10% band, and small the bottom 90%. Firm-years, gross trade, net exposure, and net/gross are customs-universe averages over 2011–2018, with net/gross the ratio of net exposure to gross trade. The *net/sales* row uses the regression sample as a reference, since sales require balance-sheet information, which is absent for many firms in the full trade sample. MPX salaries is the marginal propensity to spend on salaries out of valuation income, the size-tertile estimate of Table 4 mapped by size group (top 5% to large, 5–10% to medium, bottom 90% to small) and applied to both positions. The propensity to spend is identified from the dominant-currency coefficient in section 6 and applied here to the full USD exposure. Standard errors are in parentheses.

Table C.14: Firm-type composition of net importers and net exporters by resale intensity

	Net exporters	Net importers
Firm type (percent of firms)		
Distributor	10.1	18.1
Trader	23.2	38.4
Wholesale-type (distributor + trader)	33.3	56.5
Producer	66.7	43.5

Notes: Share of firms within each net-trade position. Firms are classified from their income statements by two ratios, each averaged over a firm's years. The first is the share of turnover that is resale of merchandise bought without transformation: merchandise sales over total net sales. The second is the share of operating costs spent purchasing goods for resale: merchandise purchases over total operating expenses. Each ratio is split at one half. *Producers* lie below the cutoff on both margins and transform inputs into their own output. *Distributors* and *traders* sell mostly resold goods and form the wholesale-type group.

Table C.15: Trade and financing by sector

	Manufacturing	Wholesale	Construction	Other
Trade (percent of sales)				
Imports/sales	9.44	22.4	5.65	6.74
median	1.10	4.66	0.22	0.00
Exports/sales	21.0	15.3	13.9	22.4
median	6.68	1.87	3.35	4.86
Liquidity and financing (percent of capital)				
Cash holdings	47.4	149	44.6	64.5
median	9.80	37.7	11.4	7.52
Debt to suppliers	88.7	330	81.5	200
median	31.9	113	36.4	42.7
Interest charged	2.54	6.97	1.97	4.37
median	0.84	1.77	0.79	1.14
Firm-years	124,258	99,973	3,035	26,158

Notes: Means with medians beneath, by macro-sector. Trade flows are percent of sales. The liquidity and financing panel is percent of fixed capital.

D Proofs and Derivations for Section 3

D.1 Proof of Lemma 1

Proof. The reset price maximizes expected discounted profits over the horizons at which the price is still expected to be in force. Ignoring terms that do not depend on the reset price, the problem is

$$\bar{p}_t^j \in \arg\max_{p^j} \sum_{h=0}^{\infty} (\beta\theta)^h \mathbb{E}_t \left[\Pi(p^j + e_{t+h}^{*/j}; \mathbf{e}_{t+h}, \boldsymbol{\eta}_{t+h}) \right]. \tag{D.1}$$

The first-order condition is

$$0 = \sum_{h=0}^{\infty} (\beta\theta)^h \mathbb{E}_t \left[\Pi_1(\bar{p}_t^j + e_{t+h}^{*/j}; \mathbf{e}_{t+h}, \boldsymbol{\eta}_{t+h}) \right]. \tag{D.2}$$

Each term in this sum is the expected marginal profit the price will generate at horizon h if it is still in force. A single reset price cannot set every term to zero, so the optimum balances expected marginal profits across horizons. Take the reference point at each horizon to be the desired price, which by definition sets the marginal profit at that horizon to zero: $\Pi_1(\cdot) = 0$. Expanding each marginal-profit term in (D.2) to first order around the corresponding desired price leaves:

$$\Pi_1(\bar{p}_t^j + e_{t+h}^{*/j}; \mathbf{e}_{t+h}, \boldsymbol{\eta}_{t+h}) \approx \Pi_{11} \left(\bar{p}_t^j - \tilde{p}_{t+h}^j \right).$$

Substituting into (D.2) and dividing by $\Pi_{11} \neq 0$,

$$0 = \sum_{h=0}^{\infty} (\beta\theta)^h \mathbb{E}_t \left[\bar{p}_t^j - \tilde{p}_{t+h}^j \right] = \frac{\bar{p}_t^j}{1 - \beta\theta} - \sum_{h=0}^{\infty} (\beta\theta)^h \mathbb{E}_t \left[\tilde{p}_{t+h}^j \right],$$

Solving for $\bar{p}_t^j$ gives the weighted average of expected desired prices in (6). □

D.2 Proof of Proposition 1

Proof. Fix $j \neq €$. Define the value wedge under the two invoicing regimes and evaluated at the same inherited buyer-currency price,

$$D_t \equiv V_j(p_{t-1}^j; \mathrm{e}_t, \boldsymbol{\eta}_t) - V_€(p_{t-1}^€; \mathbf{e}_t, \boldsymbol{\eta}_t).$$

Throughout, $o(\cdot)$ collects terms that are second order in the innovations relative to the inherited state.

First, using (4) for the two invoicing regimes and subtracting gives

$$
\begin{aligned}
D_t = {} & \Pi(p^{j}_{t-1} + e^{*/j}_t; \mathbf{e}_t, \boldsymbol{\eta}_t) - \Pi(p^{€}_{t-1} + e^{*/€}_t; \mathbf{e}_t, \boldsymbol{\eta}_t) \\
& + \beta\theta \mathbb{E}_t[D_{t+1}] + \beta(1-\theta)\mathbb{E}_t\left[\bar{V}_j(\mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1}) - \bar{V}_€(\mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1})\right]. \qquad \text{(D.3)}
\end{aligned}
$$

The wedge today is the current profit difference, plus the discounted wedge next period if the price does not reset (with probability θ), plus the discounted difference in reset values.

The reset values $\bar{V}_j$ and $\bar{V}_€$ in (5) are evaluated at optimum and therefore cancel to the first order. The direct effects of $(\mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1})$ on profits are the same for the same firm under either regime and cancel in the difference. (D.3) then becomes:

$$
D_t = \Delta\Pi_{j,t} + \beta\theta\mathbb{E}_t[D_{t+1}] + o(\|\Delta\mathbf{e}_t, \Delta\boldsymbol{\eta}_t\|),
$$

where

$$
\Delta\Pi_{j,t} \equiv \Pi(p^{j}_{t-1} + e^{*/j}_t; \mathbf{e}_t, \boldsymbol{\eta}_t) - \Pi(p^{€}_{t-1} + e^{*/€}_t; \mathbf{e}_t, \boldsymbol{\eta}_t).
$$

Solving the recursion forward gives

$$
\begin{aligned}
D_t = \sum_{h=0}^{\infty} (\beta\theta)^h \mathbb{E}_t \Big[& \Pi(p^{*}_{t-1} + \Delta_h e^{*/j}_t; \mathbf{e}_{t-1} + \Delta_h \mathbf{e}_t, \boldsymbol{\eta}_{t-1} + \Delta_h \boldsymbol{\eta}_t) \\
& - \Pi(p^{*}_{t-1} + \Delta_h e^{*/€}_t; \mathbf{e}_{t-1} + \Delta_h \mathbf{e}_t, \boldsymbol{\eta}_{t-1} + \Delta_h \boldsymbol{\eta}_t) \Big] + o(\cdot). \qquad \text{(D.4)}
\end{aligned}
$$

Each term in (D.4) compares profits at the same future date under the two regimes. A first-order expansion of each profit term around $(p^{*}_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})$ gives the same direct effects of $\Delta_h \mathbf{e}_t$ and $\Delta_h \boldsymbol{\eta}_t$ in the two regimes, and these common effects cancel in the difference. The only first-order difference left is how the inherited buyer-currency price itself has moved:

$$
\begin{aligned}
& \Pi(p^{*}_{t-1} + \Delta_h e^{*/j}_t; \mathbf{e}_{t-1} + \Delta_h \mathbf{e}_t, \boldsymbol{\eta}_{t-1} + \Delta_h \boldsymbol{\eta}_t) \\
& \quad - \Pi(p^{*}_{t-1} + \Delta_h e^{*/€}_t; \mathbf{e}_{t-1} + \Delta_h \mathbf{e}_t, \boldsymbol{\eta}_{t-1} + \Delta_h \boldsymbol{\eta}_t) \\
& = \Pi_1(p^{*}_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}) \left(\Delta_h e^{*/j}_t - \Delta_h e^{*/€}_t \right) + o(\|\Delta_h \mathbf{e}_t, \Delta_h \boldsymbol{\eta}_t\|) \\
& = \Pi_1(p^{*}_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})\, \Delta_h e^{€/j}_t + o(\|\Delta_h \mathbf{e}_t, \Delta_h \boldsymbol{\eta}_t\|). \qquad \text{(D.5)}
\end{aligned}
$$

□

D.3 The structure of Π_1

This subsection derives the valuation–quantity decomposition (13) in the main text. Write $\Pi = R - C$ with euro revenue $R = P^{€}Q(p^*)$ and euro variable cost $C = MC^{€}Q(p^*)$. All variables are evaluated at the inherited state so $R = R_{t-1}$, $C = C_{t-1}$, and $\Pi_{t-1} = R_{t-1} - C_{t-1}$. Holding marginal cost and exchange rates fixed, the euro unit value is the converted buyer-currency price, $\log P^{€} = p^* + e^{€/*}$, so $\partial \log P^{€}/\partial p^* = 1$. Demand determines quantity, $\partial \log Q/\partial p^* = \varepsilon$, and cost varies with quantity alone, $\partial \log C/\partial p^* = \varepsilon$. Hence $\partial \log R/\partial p^* = 1 + \varepsilon$ and

$$\Pi_1 = \frac{\partial \Pi}{\partial p^*} = R(1+\varepsilon) - C\,\varepsilon = R + \varepsilon(R - C) = R_{t-1} + \varepsilon\,\Pi_{t-1},$$

which is (13).

D.4 More general regime-specific exposures

The comparison in Proposition 1 assumes that the only difference between invoicing regimes is the currency in which the posted output price is written; the profit function itself is the same. In practice the two regimes may also differ in other aspects and invoice choices may *select* according to such characteristics. This subsection records what the empirical specification recovers in that case. Let

$$\widetilde{\Pi}_{j,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t), \qquad j \in \{*, €, \$\},$$

be any differentiable current-profit evaluation under invoicing regime j. This object may include output-price stickiness, cost-side currency exposure, inherited nominal obligations, treasury positions, distribution contracts, or imported-input contracts.

To first order, the wedge between regimes is linear in the shocks,

$$\widetilde{\Pi}_{j,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) - \widetilde{\Pi}_{€,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) = (\boldsymbol{\gamma}_j^e)'\,\Delta \mathbf{e}_t + (\boldsymbol{\gamma}_j^\eta)'\,\Delta \boldsymbol{\eta}_t + o(\|\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t\|), \tag{D.6}$$

with regime-specific loading vectors $\boldsymbol{\gamma}_j^e$ on the exchange-rate legs and $\boldsymbol{\gamma}_j^\eta$ on the demand shifters.

The main-text benchmark, in which the two regimes share the same profit function, is the special case

$$\widetilde{\Pi}_{j,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) = \Pi(p^*_{t-1} + \Delta e_t^{*/j}; \mathbf{e}_{t-1} + \Delta \mathbf{e}_t, \boldsymbol{\eta}_{t-1} + \Delta \boldsymbol{\eta}_t). \tag{D.7}$$

In this case, common direct effects of $\Delta \mathbf{e}_t$ and $\Delta \boldsymbol{\eta}_t$ cancel in the cross-regime difference, and the general wedge collapses to

$$\widetilde{\Pi}_{j,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) - \widetilde{\Pi}_{€,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) = \Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}) \Delta e_t^{€/j} + o(\|\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t\|). \quad \text{(D.8)}$$

Proposition 1, however, does not extend to these additional characteristics. A dollar-debt example makes the point. Let the two regimes inherit the same nominal obligation, with the same euro value at date $t-1$ rates:

$$b_{t-1} \equiv b^{€}_{t-1} = b^{\$}_{t-1} + e^{€/\$}_{t-1}, \quad \text{(D.9)}$$

where $b^{€}_{t-1}$ and $b^{\$}_{t-1}$ are log face values fixed at $t-1$ in each denomination. This is the balance-sheet analogue of the common inherited price (8): the two regimes hold the same obligation and differ only in the currency in which it is written. Suppose

$$\widetilde{\Pi}_{€,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) = \Pi(p^*_{t-1} + \Delta e_t^{*/€}; \mathbf{e}_t, \boldsymbol{\eta}_t) - \exp(b_{t-1}), \quad \text{(D.10)}$$

$$\widetilde{\Pi}_{\$,t-1}(\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t) = \Pi(p^*_{t-1} + \Delta e_t^{*/\$}; \mathbf{e}_t, \boldsymbol{\eta}_t) - \exp(b_{t-1} + \Delta e_t^{€/\$}). \quad \text{(D.11)}$$

Only the dollar-denominated obligation moves with the euro-dollar leg and the first-order wedge is now

$$\widetilde{\Pi}_{\$,t-1} - \widetilde{\Pi}_{€,t-1} = \left[\Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}) - \exp(b_{t-1})\right] \Delta e_t^{€/\$} + o(\|\Delta \mathbf{e}_t, \Delta \boldsymbol{\eta}_t\|). \quad \text{(D.12)}$$

Even if the inherited buyer-currency output price is at its envelope point, so that $\Pi_1 = 0$, the debt term would remain. In the general case, the leg-specific coefficients of (39) therefore might recover the firm's total exposure to each leg, of which the sticky-price wedge Π_1 is one component.

D.5 Derivation of the investment condition

Internal funds are operating profit Π_t^j plus the regime-common constant $\bar{n}_t$. External funds b_t^j cover the gap between investment and internal funds at cost $\mathcal{C}(b_t^j, K_t)$. The firm solves

$$\begin{aligned} V_t^j(K_t, \Pi_t^j; p^j, \mathbf{e}_t, \boldsymbol{\eta}_t) = \max_{I_t^j, b_t^j} \Big\{ & \Pi_t^j(K_t; p^j, \mathbf{e}_t, \boldsymbol{\eta}_t) - I_t^j - \mathcal{C}(b_t^j, K_t) \\ & + \beta \mathbb{E}_t \left[V_{t+1}^j(K_{t+1}, \Pi_{t+1}^j; p^j, \mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1}) \right] \Big\} \end{aligned} \quad \text{(D.13)}$$

subject to

$$K_{t+1} = (1-\delta_K)K_t + I_t^j, \qquad I_t^j = \Pi_t^j + \bar{n}_t + b_t^j.$$

Let υ_t be the multiplier on the financing constraint, the shadow value of internal funds. The first-order conditions are

$$q_t^j - 1 = \upsilon_t, \tag{D.14}$$

$$\mathcal{C}_b(b_t^j, K_t) = \upsilon_t, \tag{D.15}$$

where

$$q_t^j \equiv \beta \mathbb{E}_t \left[V_{K,t+1}^j(K_{t+1}, \Pi_{t+1}^j; p^j, \mathbf{e}_{t+1}, \boldsymbol{\eta}_{t+1}) \right].$$

Here $V_{K,t+1}^j$ is the *total* marginal value of capital. It includes both the direct effect of K_{t+1} on continuation payoffs and its indirect effect through next-period operating profit $\Pi_{t+1}^j(K_{t+1}; \cdot)$.

The two first-order conditions say that the firm invests until the marginal value of capital, net of its unit cost, equals the shadow value of internal funds, and borrows until the marginal cost of external funds equals that same shadow value. Combining them and using $b_t^j = I_t^j - \Pi_t^j - \bar{n}_t$ gives

$$q_t^j - 1 = \mathcal{C}_b(I_t^j - \Pi_t^j - \bar{n}_t, K_t).$$

Differentiating:

$$\frac{\partial I_t^j}{\partial q_t^j} = \frac{1}{\mathcal{C}_{bb}(b_t^j, K_t)}, \qquad \frac{\partial I_t^j}{\partial \Pi_t^j} = 1.$$

The second derivative says that, at a given marginal q, an extra euro of operating profit substitutes one for one for costly external funds and passes into investment. Taking differences in this linearized condition across the j- and euro-invoicing counterfactuals gives (29).

D.6 The marginal-q response to the exchange rate

The closed form (D.17) follows from applying the inherited-price logic of Proposition 1 to the scale derivative Π_{1K}. Holding non-exchange-rate shocks fixed, the marginal-q response is the discounted stream of future scale-dependent profit effects generated while the inherited price remains in force,

$$\Delta q_t^{j-€} \approx \sum_{h=1}^{\infty} (\beta\theta)^h \Pi_{1K}(p_{t-1}^*; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}, K_t) \mathbb{E}_t[\Delta_h e_t^{€/j}]. \tag{D.16}$$

The sum starts at $h = 1$, not $h = 0$. Under the random-walk benchmark (9), $\mathbb{E}_t[\Delta_h e_t^{€/j}] = \Delta e_t^{€/j}$ for every h, and the geometric sum collapses to

$$\Delta q_t^{j-€} \approx \frac{\beta\theta}{1-\beta\theta} \Pi_{1K}(p_{t-1}^*; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1}, K_t) \Delta e_t^{€/j}. \tag{D.17}$$

D.7 Short-maturity hedging

This subsection derives the residual value wedge (31) under maximal short-maturity hedging. Because the rest of the model uses $e_t^{€/j}$ for the log exchange rate, denote its level counterpart by

$$E_t^{€/j} \equiv \exp(e_t^{€/j}), \tag{D.18}$$

the number of euros per unit of currency j. Let $F_{s,s+\ell}^{€/j}$ denote the level forward rate agreed at date s for delivery at $s + \ell$. We use s for the generic hedge-initiation date, reserving t for the date of the exchange-rate shock.

If at date s the firm sells $H_{s,s+\ell}^j$ units of currency j forward, the euro payoff at settlement is

$$\varphi_{s,s+\ell} = H_{s,s+\ell}^j \left(F_{s,s+\ell}^{€/j} - E_{s+\ell}^{€/j} \right). \tag{D.19}$$

Thus, if the firm has a known foreign-currency receipt $X_{s+\ell}^j$ at date $s + \ell$ and fully hedges it by setting $H_{s,s+\ell}^j = X_{s+\ell}^j$, the combined euro cash flow is

$$X_{s+\ell}^j E_{s+\ell}^{€/j} + X_{s+\ell}^j \left(F_{s,s+\ell}^{€/j} - E_{s+\ell}^{€/j} \right) = X_{s+\ell}^j F_{s,s+\ell}^{€/j}. \tag{D.20}$$

The hedge therefore fixes the euro value of that particular receipt at the forward rate agreed when the hedge is written.

Assume, for simplicity, that the log forward rate equals the conditional expectation of the future log spot rate:

$$\log F_{s,s+\ell}^{€/j} = \mathbb{E}_s \left[e_{s+\ell}^{€/j} \right]. \tag{D.21}$$

Under the random-walk benchmark for exchange rates,

$$e_{t+h}^{€/j} = e_t^{€/j} + \varepsilon_{t+1}^{€/j} + \cdots + \varepsilon_{t+h}^{€/j}, \qquad \mathbb{E}_t[\varepsilon_{t+h}^{€/j}] = 0, \tag{D.22}$$

this implies

$$\log F_{s,s+\ell}^{€/j} = e_s^{€/j}. \tag{D.23}$$

Now consider the date-t exchange-rate innovation $\Delta e_t^{€/j} \equiv e_t^{€/j} - e_{t-1}^{€/j}$. Suppose it arrives after all date-$t-1$ hedge positions have been chosen. If the maximum hedge maturity is

L, pre-shock hedges can cover only cash flows settling at horizons $h = 0, \ldots, L-1$. In a maximal-hedging benchmark (all transactions are always fully hedged), these already-hedged contractual cash flows are insulated from the date-t innovation:

$$\Delta\Pi^{j-€}_{H,t+h} = 0, \qquad h = 0, \ldots, L-1, \tag{D.24}$$

where $\Delta\Pi^{j-€}_{H,t+h}$ denotes the hedged profit response at horizon h from invoicing in currency j rather than in euros.[30]

For horizons $h \geq L$, the cash flow at date $t+h$ can be hedged no earlier than date $t+h-L$, when the date-t shock has already occurred. The relevant forward rate therefore incorporates the post-shock exchange-rate level:

$$\log F^{€/j}_{t+h-L,t+h} = e^{€/j}_{t+h-L}. \tag{D.25}$$

Under the random-walk benchmark, a hedge written at $t+h-L$ removes only the exchange-rate innovations realized after $t+h-L$, not the level shift generated by $\Delta e^{€/j}_t$. Therefore, conditional on the inherited price still being in force, the horizon-h profit response remains

$$\Delta\Pi^{j-€}_{H,t+h} \approx \Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})\Delta e^{€/j}_t, \qquad h \geq L. \tag{D.26}$$

Combining (D.24) and (D.26), the residual value wedge after all short-dated hedgeable transactions are hedged is

$$\begin{aligned} V^{j-€}_t(L) &\equiv \sum_{h=0}^{\infty}(\beta\theta)^h \mathbb{E}_t\left[\Delta\Pi^{j-€}_{H,t+h}\right] \\ &\approx \sum_{h=L}^{\infty}(\beta\theta)^h \Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})\Delta e^{€/j}_t = \frac{(\beta\theta)^L}{1-\beta\theta}\Pi_1(p^*_{t-1}; \mathbf{e}_{t-1}, \boldsymbol{\eta}_{t-1})\Delta e^{€/j}_t, \end{aligned} \tag{D.27}$$

which is (31).

[30] This benchmark is the best case for hedging. In practice, some components of Π_1, such as quantity responses or uncertain future sales, may be harder to hedge than known contractual receipts.